Comparison on LCC- and MMC- HVDC Protection Design - A Review

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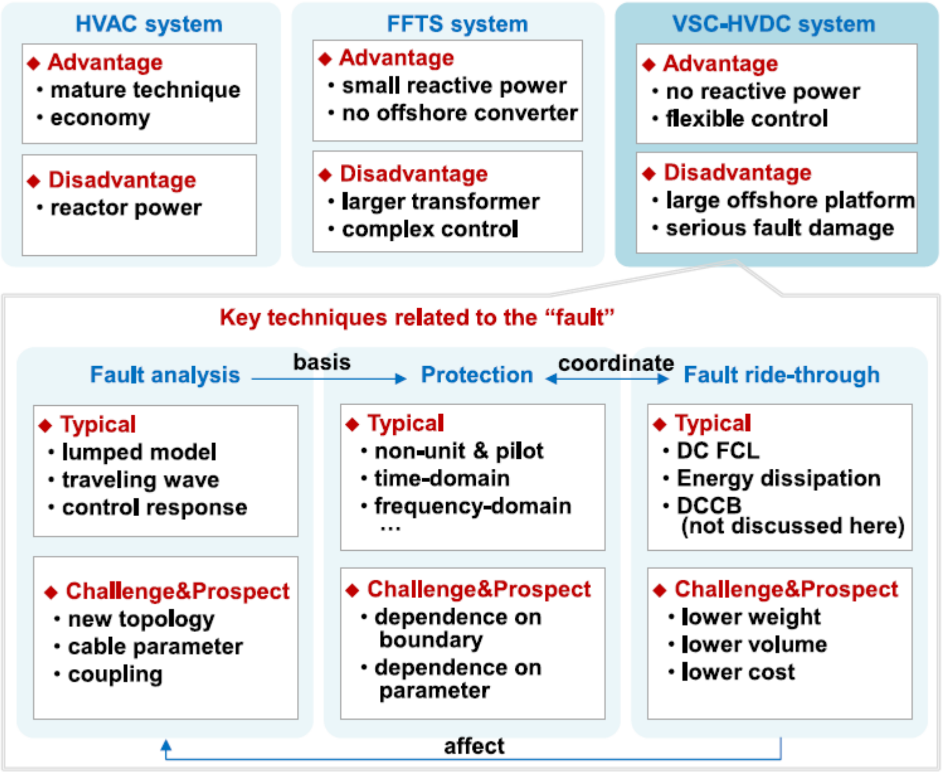
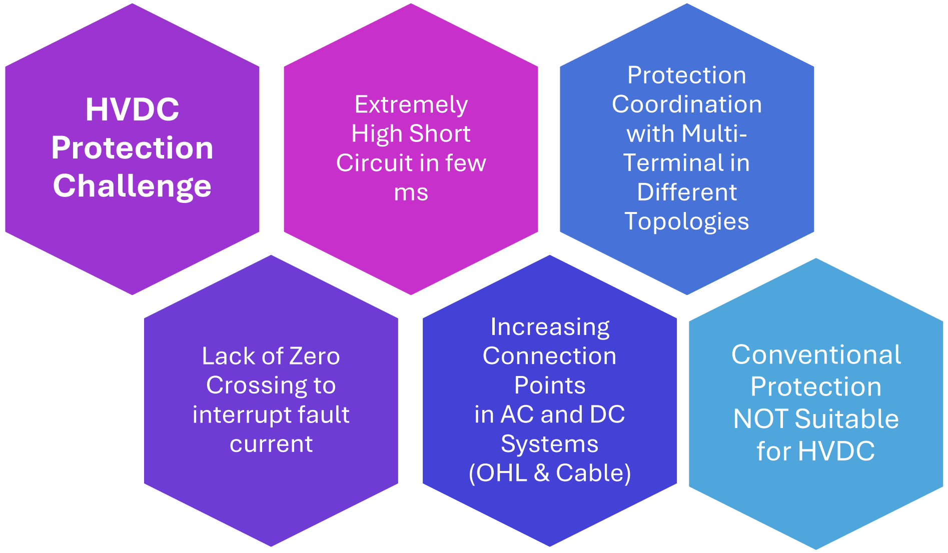
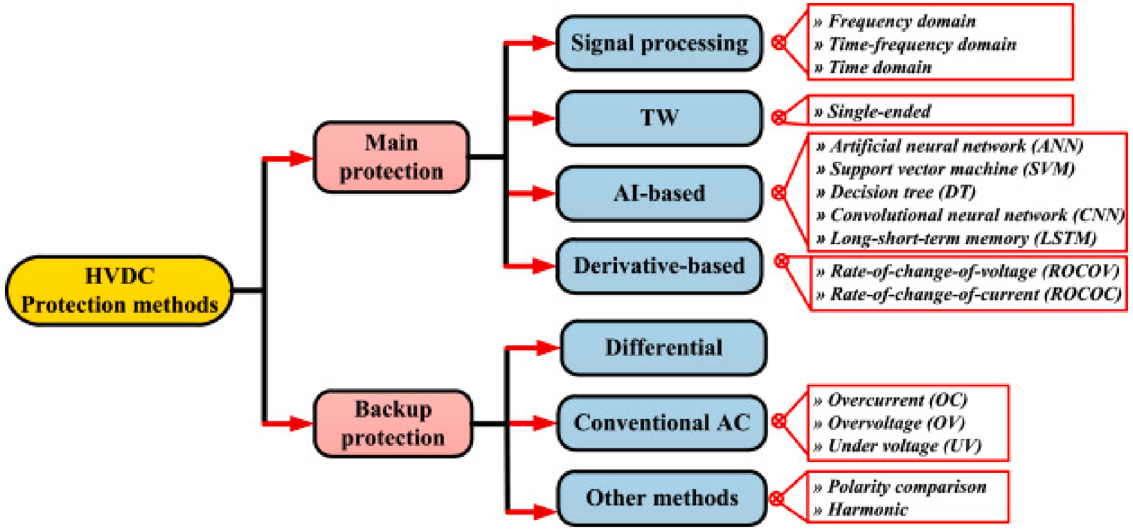
Considerations in HVDC Protection?

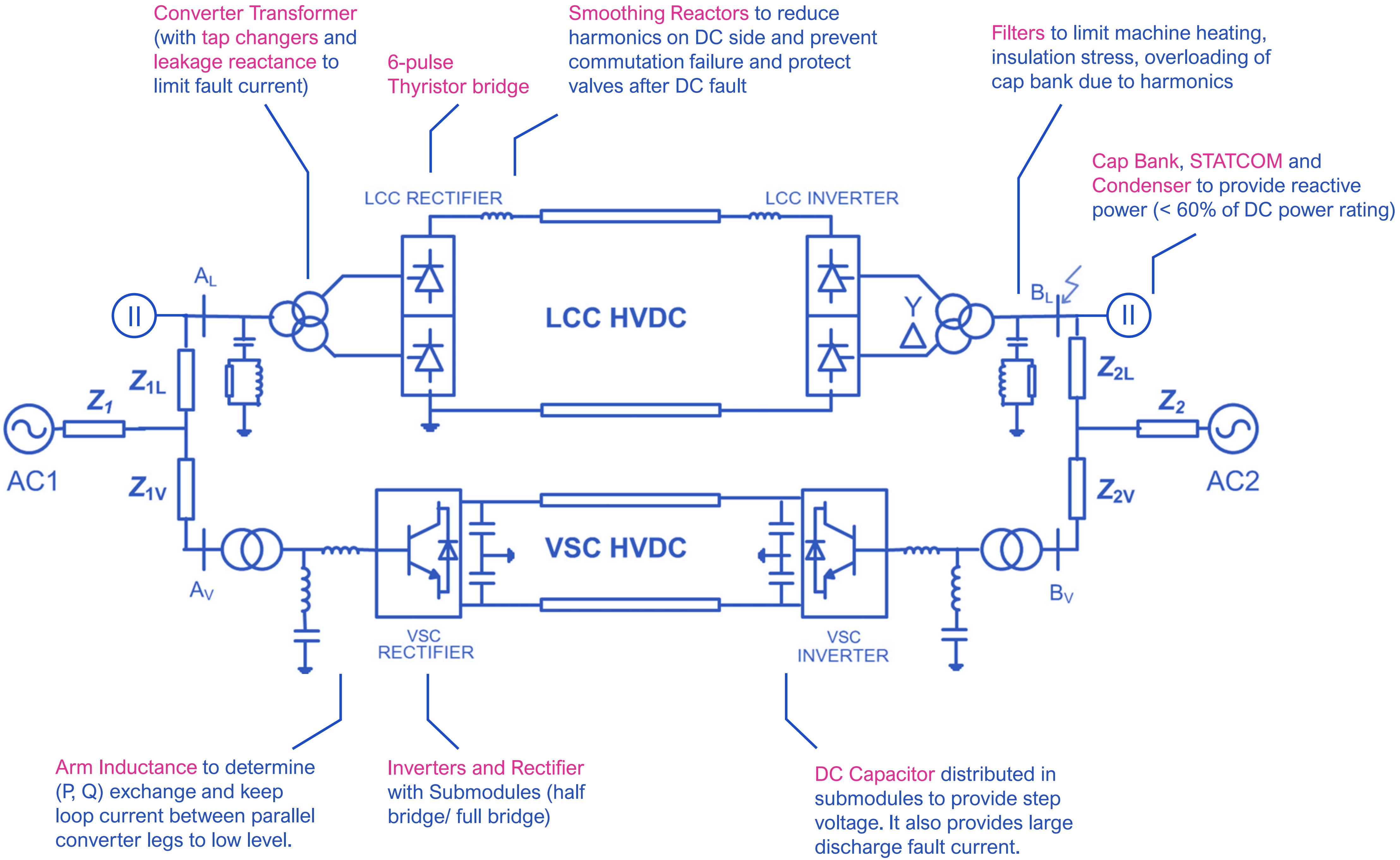




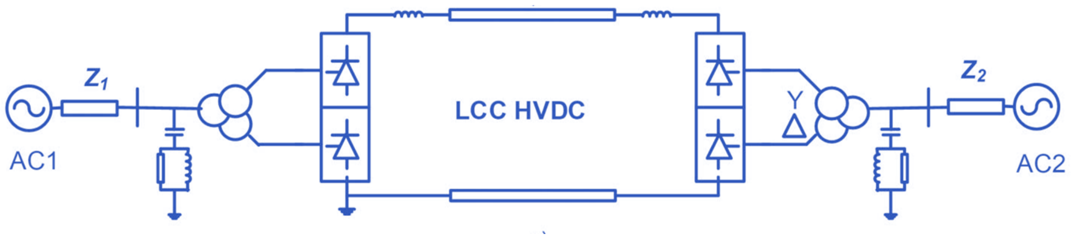


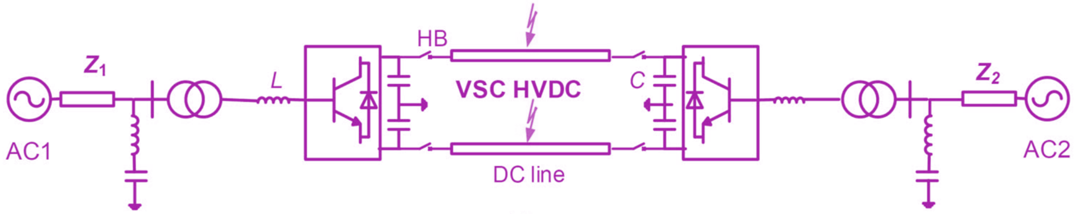
Content

- Introduction on HVDC – LCC vs MMC (VSC)
- Protection Philosophy
- Fault Interruption
    - Fault Interruption – AC and DC
    - HVDC Circuit Breakers
    - DC Fault Clearance – Converter Response (HB MMC vs FB MMC)
    - Fault Current Limiter (FCL)
- HVDC Protection in General
    - Control Action
    - HVDC Line Protection
    - Fault Detection, Sectionalizing, Location and Gate Shutting
- Protection – LCC HVDC
    - Converter Protection Zone Failure
        - Commutation Failure
        - Valve Failure – Misfire, Arc Through and Current Extinction
- Protection – MMC HVDC
    - MMC HVDC Protection in Different Zones
    - Fault Capability and Ride-through


Converter Transformer (with tap changers and leakage reactance to limit fault current)
6-pulse Thyristor bridge
Smoothing Reactors to reduce harmonics on DC side and prevent commutation failure and protect valves after DC fault
Filters to limit machine heating, insulation stress, overloading of cap bank due to harmonics
Cap Bank, STATCOM and Condenser to provide reactive power (< 60% of DC power rating)
LCC RECTIFIER
LCC INVERTER
LCC HVDC
A_L
B_L
Y
Δ
Z_1L
Z_1
Z_2L
Z_2
AC1
AC2
Z_1V
Z_2V
VSC HVDC
A_V
B_V
VSC RECTIFIER
VSC INVERTER
Arm Inductance to determine (P, Q) exchange and keep loop current between parallel converter legs to low level.
Inverters and Rectifier with Submodules (half bridge/ full bridge)
DC Capacitor distributed in submodules to provide step voltage. It also provides large discharge fault current.

Technical Challenges towards LCC – HVDC and VSC – HVDC





Traditionally HVDC grid were realized as LCC as a current source converter (CSC), but was practically infeasible with the following reason.

LCC

- Difficult current reversal
- Dependent on HVAC system strength for whole system recovery
- Need for DC and AC Filters with harmonics
- Prone to Commutation Failure
- Only works when inverter is in active state

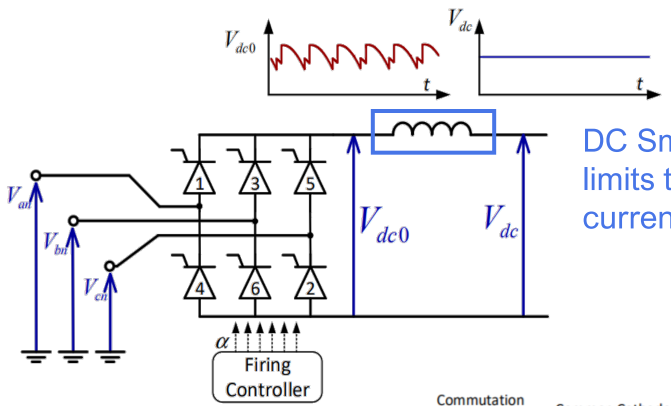
With the advent of IGBTs, VSC (Particularly MMC type) for HVDC system was introduced to overcome drawbacks for LCC. Yet, there are new technical challenge.

VSC (MMC)

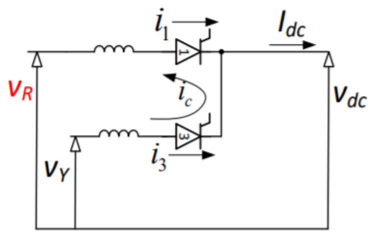
- Protection Coordination
- Fast Fault Removal with Fault Location
- Sophisticated DC Voltage Control & Power Flow
- Functions during Communication Failure
- Load Supply after Isolating a Section of HVDC grid

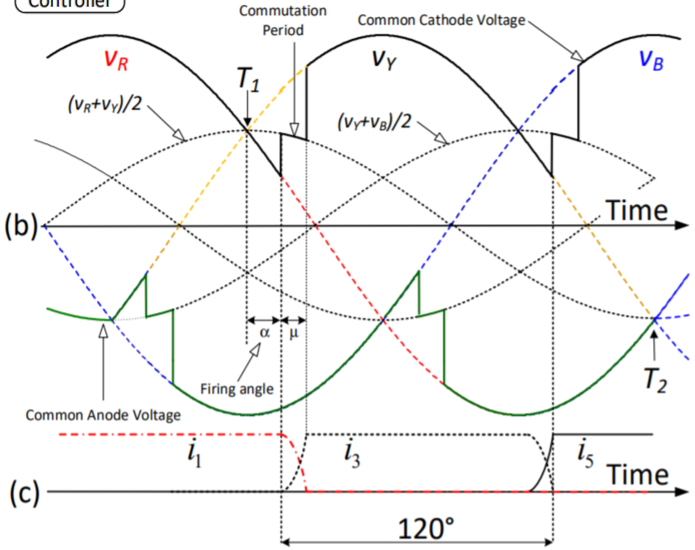
Comparison between LCC – HVDC and VSC – HVDC

LCC – HVDC



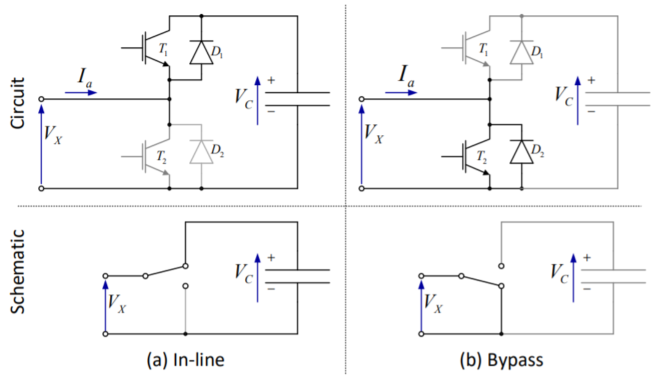
DC Smoothing Reactor limits the rise of fault current





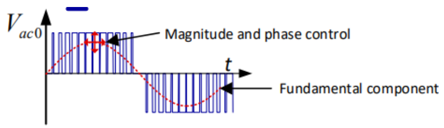
1D Control – Firing Angle α at Rectifier End and Extinction Angle γ at Inverter End

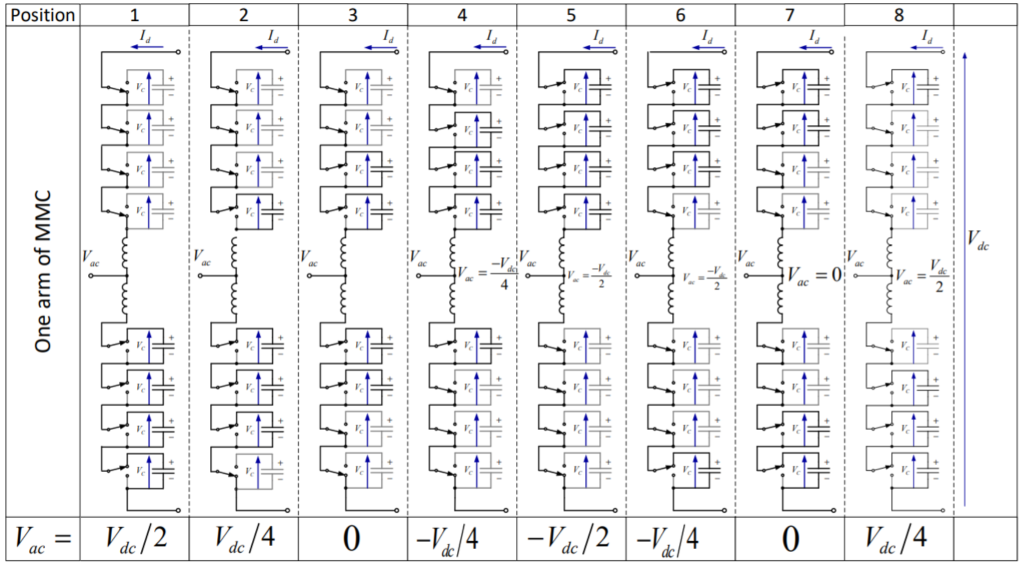
VSC – HVDC



Capacitor discharges rapidly

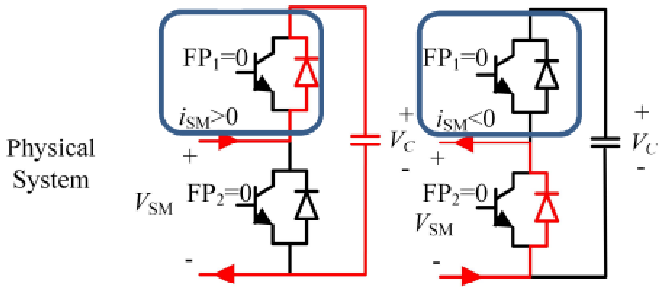
No commutation failure, but incapable of limit overvoltage and current



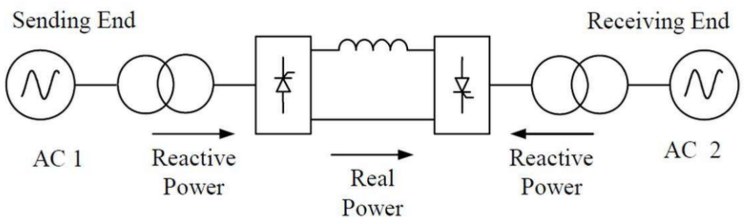


2D Control – Converter Voltage U_C

Power Flow in LCC – HVDC and VSC - HVDC



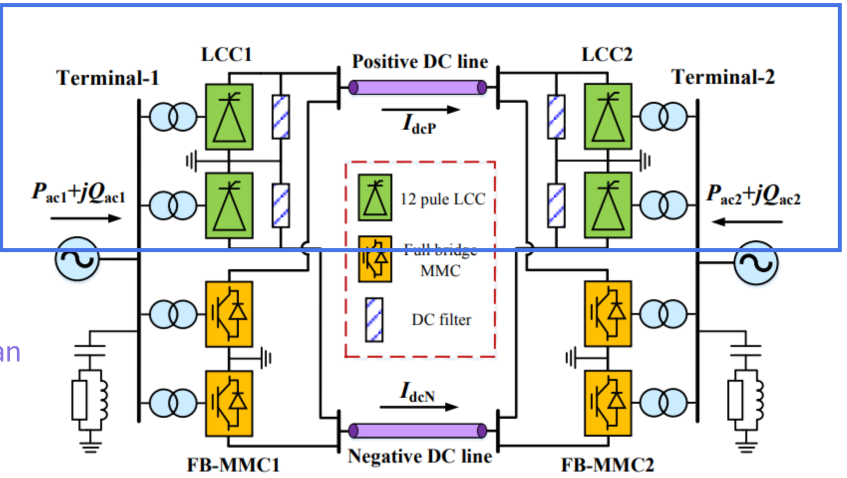
LCC – HVDC



$$P_{\mathrm{LCC}} = (V_{\mathrm{dc0}} \cos\alpha - d_x I_{\mathrm{dcLCC}}) I_{\mathrm{dcLCC}}$$

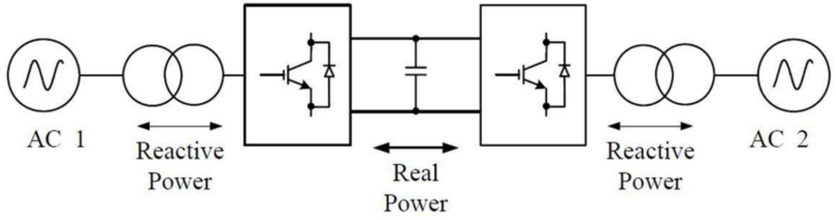
$$Q_{\mathrm{LCC}} = V_{\mathrm{dc0}} I_{\mathrm{dcLCC}} \sqrt{1 - \left(\cos\alpha - \frac{d_x I_{\mathrm{dcLCC}}}{V_{\mathrm{dc0}}}\right)^2}$$

Require large reactive power



VSC – HVDC can transmit Q for LCC – HVDC

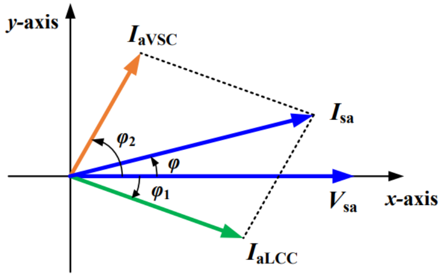
VSC – HVDC



$$P_{\mathrm{VSC}} = 1.5 u_{\mathrm{sd}} i_{\mathrm{vd}}$$

$$Q_{\mathrm{VSC}} = -1.5 u_{\mathrm{sd}} i_{\mathrm{vq}}$$

Given that d-axis is aligned with AC voltage, so u_{sd} = AC voltage and u_{sq} = 0

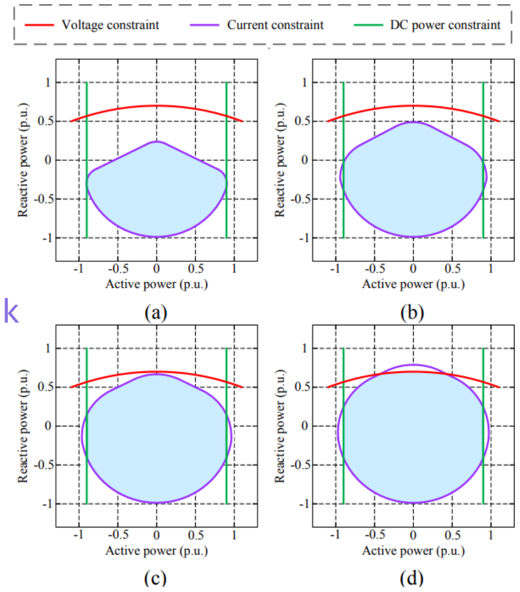


$$P^2 + Q^2 = \left(\frac{3}{2} V_{\mathrm{sa}} I_{\mathrm{sa}}\right)^2$$

$$= (P_{\mathrm{VSC}} + P_{\mathrm{LCC}})^2 + (Q_{\mathrm{VSC}} - Q_{\mathrm{LCC}})^2$$

$$k = \frac{S_{\mathrm{VSC}}}{S_{\mathrm{LCC}}}$$

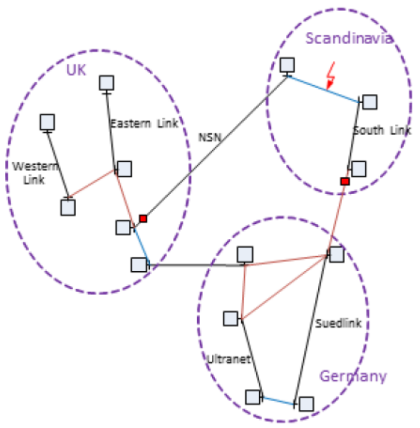
↑ k

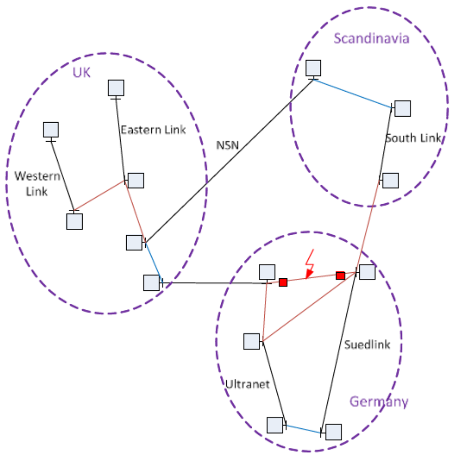


Protection Philosophy for HVDC Protection

- Protection for HVDC should consider all possible faults and abnormal running conditions (e.g. pole balance, fault ride-through at AC side, acceptable failure in sub-module).
- Overlapping protection zone should be provided without any blind spot.
- Protection for HVDC should adapt to different operating conditions (e.g. black-start, STATCOM mode, emergency power control, reverse power) without activation.
- Protection for HVDC should employ independent power source, equipment, control point and trip coil if any.
- A fast operated main protection and a backup protection with different operating mechanism should be provided to each fault type (e.g. open pole, pole-to-earth fault, pole-to-pole fault in line protection)
- The main protection for HVDC should not rely on communication between two converter substations. The backup protection, if not necessary, should also not rely on communication between two converter substations. (e.g. Travelling Wave and Voltage Change in DC Line Protection)
- Protection systems and control systems should be physically and electrically separated with independent inputs and outputs.
- Protection systems and control systems should be coordinated appropriately.
- Protection systems for DC side and that of AC side should be well coordinated. During the fault recovery stage at AC side, the DC side protection should maintain stable.
- Protection for the two poles should be independent. Protection output for a pole should not activate any converter control and breaker action at another pole.
- Avoid pole isolation at both poles under any single pole fault.

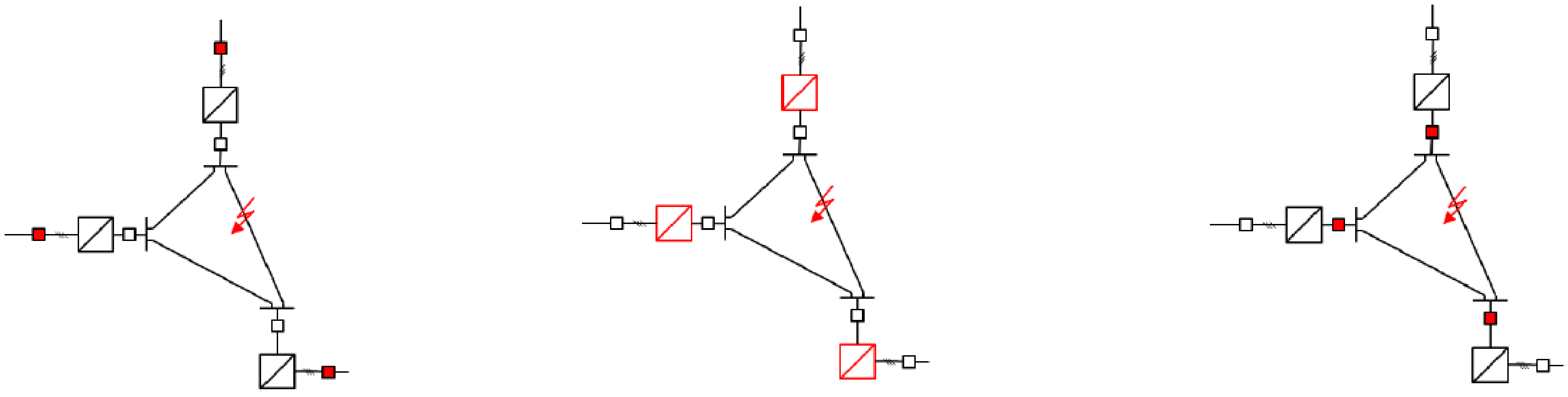
Partially Selective vs Fully Selective Fault Clearance





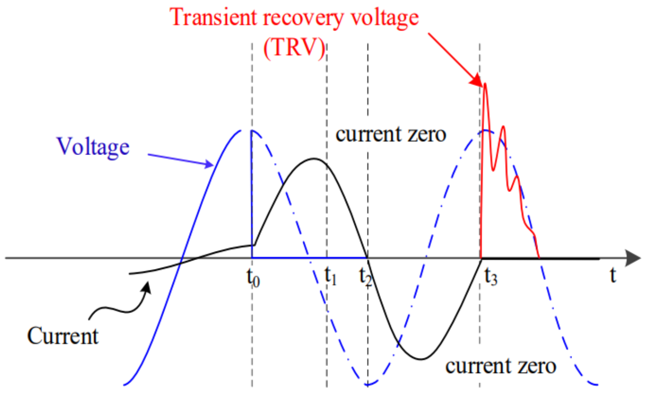
Feature	Partially Selective Fault Clearing	Fully Selective Fault Clearing
Protection Zones	Divides grid into sub-grids (e.g., by region or subsystem).	Defines zones for **each branch/node** (individual line protection).
Isolation Device	- HVDC CBs - Fault current limiters (e.g., SFCL) - DC/DC converters	HVDC CBs **at both ends of every line**.
Fault Impact	Limited to one sub-grid; healthy zones remain operational.	Limited to **only the faulted line**; rest of grid unaffected.
Equipment Requirements	Fewer HVDC CBs (only at sub-grid boundaries).	Maximum number of HVDC CBs (every line).
Complexity	Moderate (requires sub-grid coordination).	High (needs precise fault detection + backup protection).
Best For	Grids with clear subsystems (e.g., multi-terminal clusters).	**Any HVDC grid** (especially meshed networks).
Advantages	- Lower cost (fewer CBs). - Balances selectivity and simplicity.	- Minimal disruption. - Maximizes grid availability.
Disadvantages	- Partial outages still occur. - Sub-grid design critical.	- High cost (many CBs). - Complex coordination.

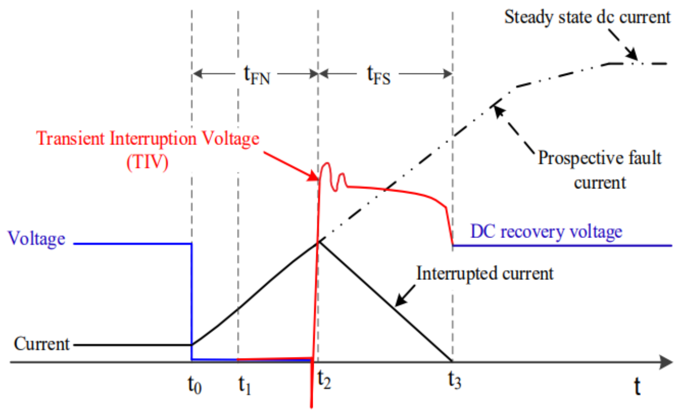
Non-Selective Fault Clearance



Aspect	AC Breakers (Scheme 1)	Fault-Blocking Converters (Scheme 2)	HVDC Circuit Breakers (Scheme 3)
How It Works	Trips AC breakers to de-energize the entire HVDC grid.	Converters block DC fault current (e.g., full-bridge MMC).	DC breakers at converter terminals interrupt fault current.
Speed	Slow (several AC cycles).	Faster (quick converter-controlled interruption).	Fastest (1–5 ms interruption).
Grid Impact	Entire HVDC grid shuts down.	Faulted line isolated quickly; partial outage.	Minimal disruption (healthy parts stay energized).
Equipment Needed	Existing AC breakers (no extra cost).	Advanced converters (e.g., full-bridge MMC).	Dedicated HVDC circuit breakers (expensive).
Pros	- No additional DC equipment needed.	- Fast fault clearing. - Supports AC grid with reactive power during faults.	- Fast isolation. - Reactive power support maintained.
Cons	- Long outage (full grid restart). - No reactive power support during faults.	- Expensive converter design. - Limited to certain topologies.	- High cost of DC breakers. - Complex to coordinate.
Best For	Small HVDC grids with low reliability needs.	Medium/large grids needing faster recovery.	Large or meshed grids requiring minimal downtime.

Fault Current Interruption





Parameter	HVAC Systems	HVDC Systems
Natural Current Zero	✅ Available (AC cycles enable interruption)	❌ Absent (requires forced commutation)
Concept	Cleared by Interruption	Cleared by Commutation
Peak Current Limitation	Inductance (L) dominates ($I_{pk} = V/X_L$)	Resistance (R) dominates ($I_{pk} = V/R$)
Fault Current Rise Time	~10–20 ms, depending on fault type & inception angle (AC time constant $\tau = L/R$)	~1–5 ms (DC time constant $\tau = L/R$, but L is small)
Interruption Mechanism	Exploits natural current zero crossings	Requires active methods (e.g., MOVs, counter-voltage to absorb ~MJ)
Critical Voltage	Transient Recovery Voltage (TRV) imposed by network	Transient Interruption Voltage (TIV) is generated by CB / Converter
Breaker Response Time	80–100 ms (standard AC CBs)	1–10 ms (ultra-fast solid-state/hybrid CBs)
Energy Absorption	Minimal (energy dissipates naturally)	Significant (MOVs absorb $\frac{1}{2}LI^2$)
Protection Schemes	Overcurrent/distance relays	di/dt sensing, hybrid CBs, current limiters

Condition under AC Faults vs DC Faults

AC fault:

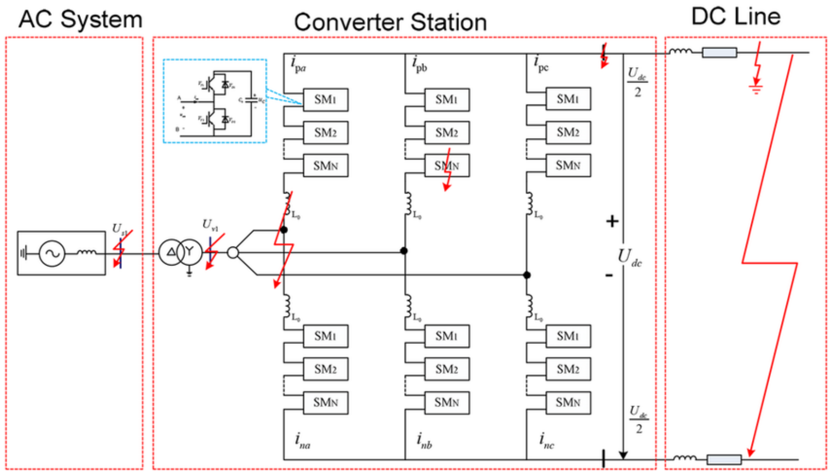
- Voltage drops at PCC and the converter supplies a voltage dependent reactive current to stabilize the voltage.
- Converter is capable to supply short circuit current up to the rating of the converter until it changes to another state (e.g. current limiting).

DC fault:

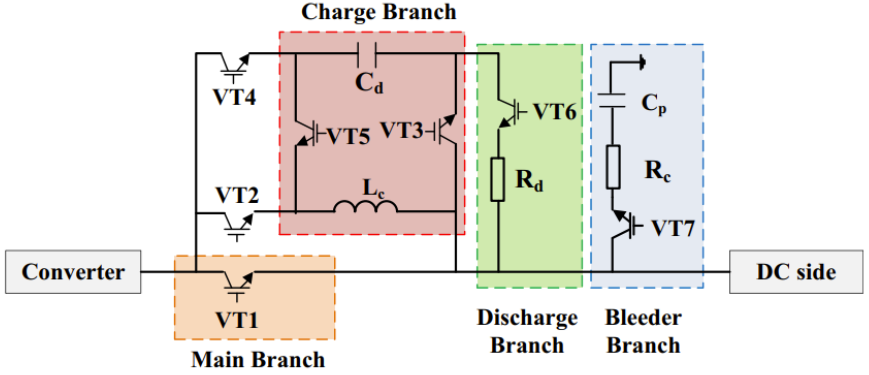
- DC Fault is harmful with large capacitor discharge current with large rate-of-rise.
- Converter Action: Fault current blocking time depends on submodule topology. Yet, fast blocking means difficulties in fault location.
- In MMC-HVDC with Half-bridge (HB) SM, IGBT is blocked with internal thyristor in the SM and AC CBs at both end must interrupt fault current. It is detrimental to converter.
- In MMC-HVDC with Full-bridge (FB) SM, fault current can be controlled like LCC based system.

Note: Some VSC designs require complete discharge of capacitors before the converter can change the operating state. It can lead to a shutdown of 90 min.

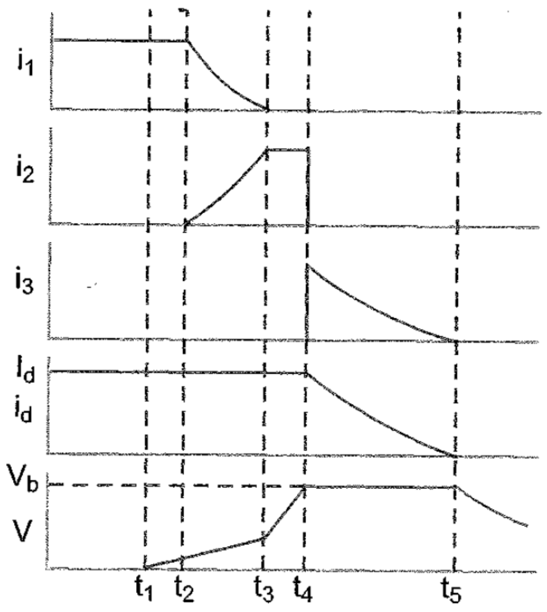
[1] Research on fault clearing scheme for half-bridge modular multilevel converters high voltage DC based on overhead transmission lines



Fault Clearance Module



Theory of HVDC Circuit Breaker



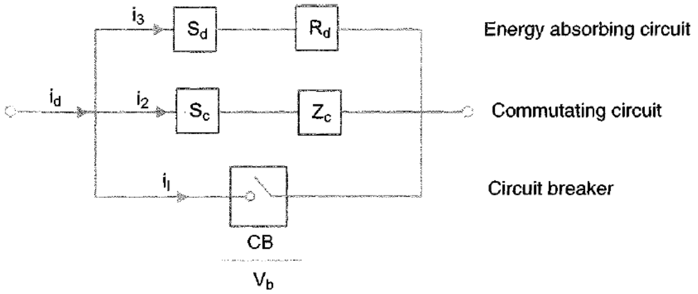
Process of HVDC CB Operation

1. **Initial State**: Current flows through closed breaker contacts (vacuum, oil, airblast, or SF_6).
2. **Trip Signal**: Contacts open at t_1, creating an arc.
3. **Commutation Circuit Activation** (at t_2):
    1. A **series L-C circuit** (Zc) is inserted via a switch (Sc, e.g., triggered gap or solid connection).
    2. This creates a **current zero-crossing**, transferring current to Zc.
4. **Energy Absorption** (at t_3):
    1. When capacitor voltage reaches Vb, nonlinear resistor (Rd) is engaged via Sd.
    2. Current decays to zero as energy dissipates in Rd.
5. **Completion**: By t_4, the breaker isolates the fault. Fast isolators may be used if residual voltage risks breaker overstress.

Energy Absorption & Current Interruption

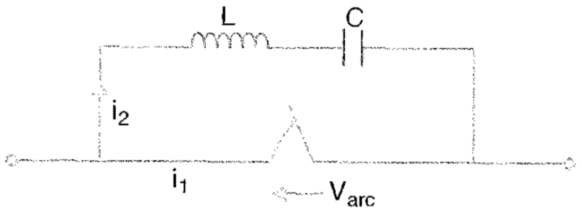
$$W_b = \frac{1}{2} L I_d^2 \left(V_b/(V_b - V_d)\right) \qquad T_i = L I_d/(V_b - V_d)$$

- The energy absorbed by the breaker (Wb) and the time to interrupt current (Ti) depend on:
    - Inductance (L) and initial current (Id).
    - Breaker contact voltage (Vb) and system voltage (Vd).
- Reducing Id (via converter current control) decreases energy absorption, lowering breaker cost.
- Fast interruption increases Ti and breaker cost, while controlled current reduction (30–50 ms) optimizes performance.



Passive Commutation Method

- An **L-C circuit** across the breaker initiates commutation via the arc itself.
- **Air blast breakers** outperform SF_6 in field tests.
- The process is governed by a differential equation describing current i_2 in the commutation loop.



$$L\left(d^2 i_2/dt^2\right) + \left(dV_{arc}/di_1\right)\left(di_2/dt\right) + (1/C) i_2 = 0$$

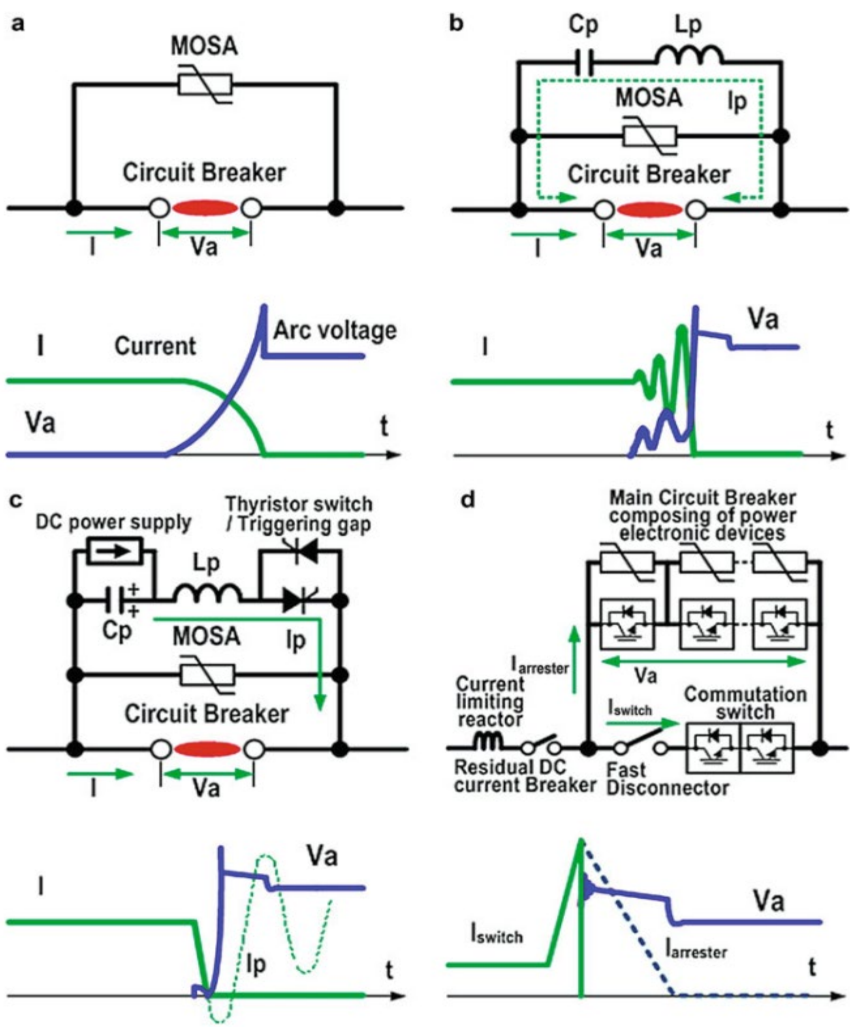
HVDC – Circuit Breaker

Design Criteria for HVDC Circuit Breaker

- Fault Current Interruption Time: should be less than 5 ms.
- Maximum Current Breaking Capability: Must handle high fault currents without damage.
- Overvoltage Protection: Must protect against voltage spikes.
- Switching Losses: Should be minimized.
- Fault Current Rate Limiting Reactor: Helps in fault detection and isolation.
- Maximum Dissipated Energy: Should be minimized to avoid large arresters.
- Total Cost: Should be kept low considering the high number of components required.

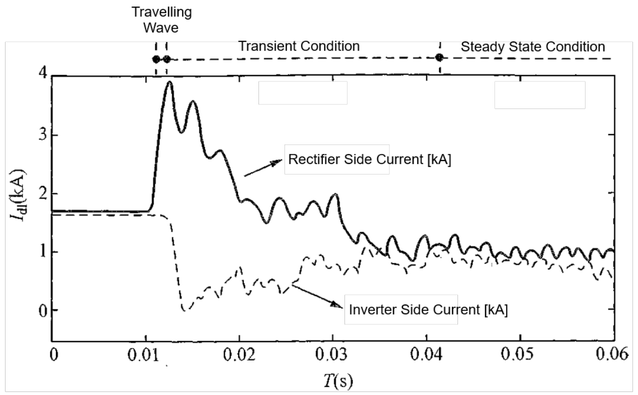
Type of HVDC Circuit Breaker

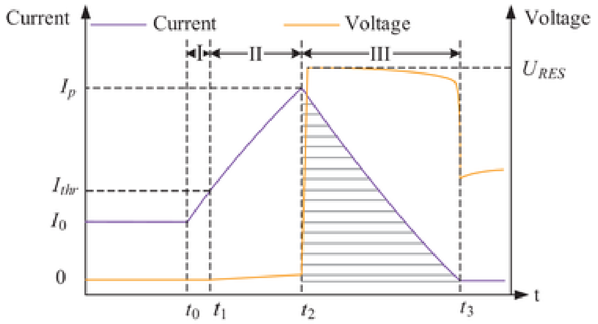
- Passive DC Circuit Breakers: Use a resonant circuit to create a current zero crossing, allowing the breaker to interrupt the current. They consist of a low-resistance mechanical interrupter, a current injection path with an LC resonant circuit, and an energy absorption branch with surge arrestors.
- Active DC Circuit Breakers: Similar to passive breakers but include a pre-charged capacitor in the auxiliary circuit to create current zero crossing when the fault current is high.
- Solid State DC circuit Breakers: Utilize IGBTs to interrupt the fault current without needing a current zero crossing. They are faster than mechanical breakers but have higher power losses and costs due to the need for multiple series-connected switches.
- Hybrid DCCB: Combine mechanical and solid-state technologies to achieve fast operation and low power losses. They have a low impedance path for normal operation and redirect current to a solid-state HV valve during a fault. This type of breaker is designed to interrupt the fault current quickly while minimizing power losses.



(a) Arc Voltage Current Limiting Scheme
(b) Passive Resonance Current Zero Creation Scheme
(c) Active Resonance Current Zero Creation Scheme with Current Injection
(d) Hybrid Mechanical and Power Electronics Switch

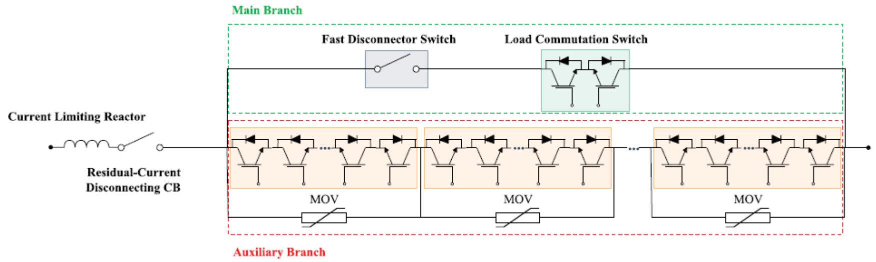
HVDC – Circuit Breaker

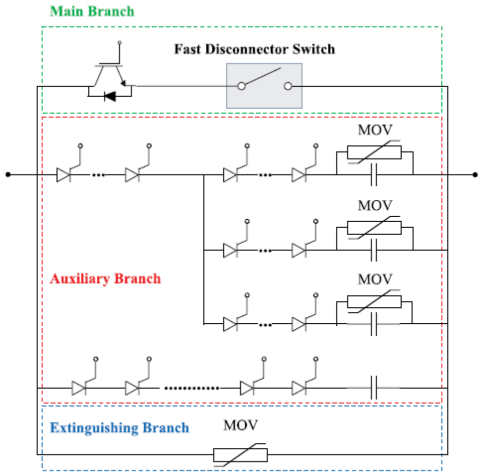


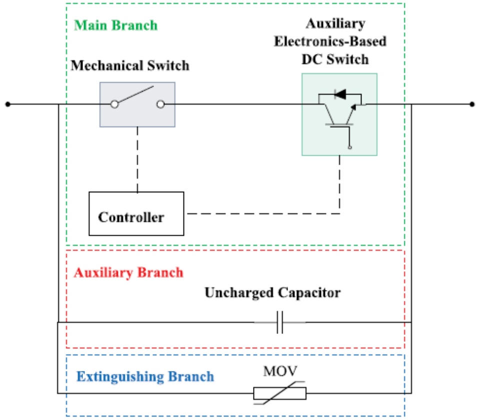


Feature	LCC-HVDC Circuit Breakers	VSC-HVDC Circuit Breakers
Type of CB	Mechanical CBs (with resonant circuits)	Solid-state or Hybrid CBs (using IGBTs, diodes)
Operating Principle	Creates artificial current zero crossing	• **Interrupt** - Create a current zero in the main path • **Commute** – Push current in path with counter voltage • **Absorb** - Limit and sustain counter voltage with arrester
Response Time	Slow (~30–50 ms)	Very fast (~1–5 ms)
Fault Current Handling	High fault current capability	Lower fault current tolerance (requires fast CB action)
Complexity	Simpler, but requires auxiliary circuits	More complex (semiconductor-based switching)
Cost	Lower initial cost, but higher maintenance	Higher initial cost, lower maintenance
Applications	Point-to-point HVDC (less critical timing)	Multi-terminal HVDC (fast isolation needed)
Fault Current Limitation	Relies on AC-side breakers (slower)	Can actively limit fault current (faster)
Key Challenges	Slow interruption, mechanical wear	High losses (for solid-state), cost, complexity

HVDC – Circuit Breaker

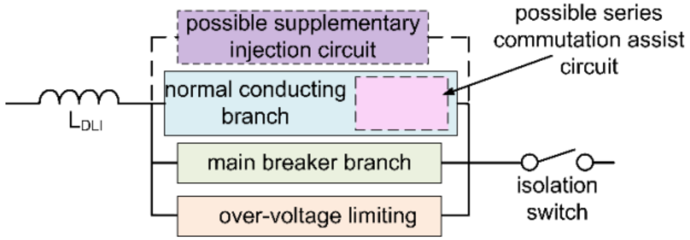






Feature	ABB HVDC CB	Alstom HVDC CB	Siemens HVDC CB
Design Type	Hybrid (Mechanical + Power Electronics)	Hybrid (Thyristor-based + Mechanical)	Hybrid (Mechanical + Power Electronics)
Key Components	- Ultrafast Disconnector (UFD) - Power electronic switches (IGBTs) - MOV surge arresters	- Thyristor stacks - Pre-charged capacitors - MOV surge arresters	- Power electronic switches - Uncharged capacitor (auxiliary branch) - Damping resistors
Operation Speed	<5 ms (fault clearance)	<5.5 ms (tested at 5,200 A, 160 kV)	Comparable to ABB (~5 ms)
Current Interruption	Uses UFD to create artificial current zero	Thyristor commutation + resonant circuits	Capacitor-based commutation
Fault Current Handling	High (tested up to 9 kA in prototypes)	High (5.2 kA interrupted in tests)	High (exact values not specified)
Conduction Losses	Negligible (main path via low-loss UFD)	Moderate (thyristor conduction losses)	Low (optimized hybrid design)
Unique Features	- Ultra-fast mechanical disconnector - Modular scalability	- SF6-free design - Mechatronic commutation	- Uncharged capacitor for commutation - Integrated damping
Applications	Multi-terminal HVDC grids	Meshed HVDC grids (e.g., offshore wind)	Point-to-point & multi-terminal HVDC

HVDC Circuit Breaker Applications

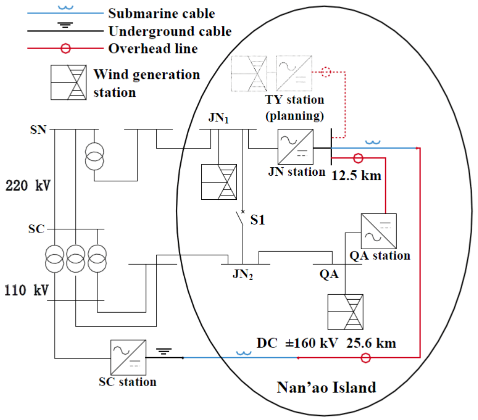


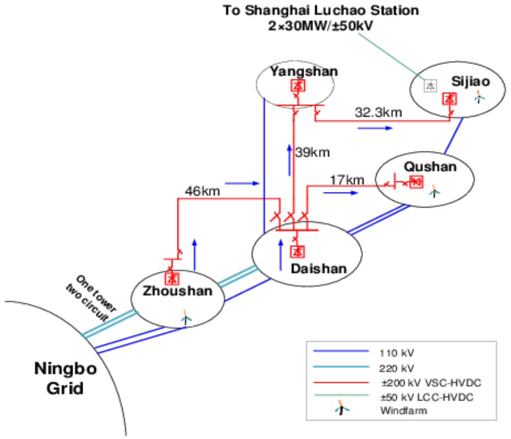
160kV Active Current Injection HVDC CB at Nan'ao Project (2013)

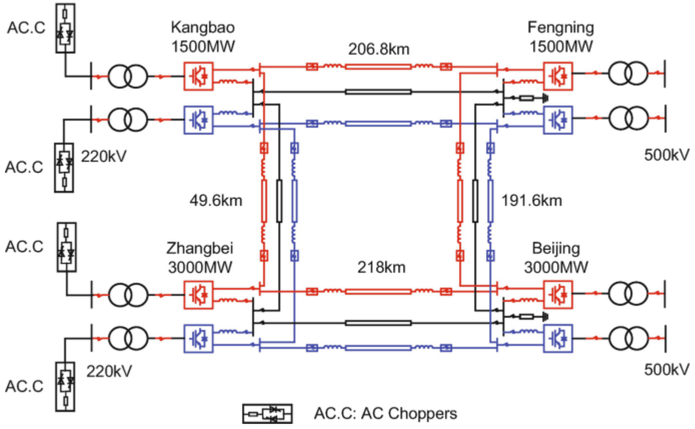
200kV Hybrid HVDC CB at Zhoushan Project (2014)

500kV Hybrid HVDC CB at Zhangbei Project (2019)

500kV Coupled Negative Voltage HVDC CB at Zhangbei Project (2019)







HVDC Circuit Breaker Applications – Zhoushan Project (2014)

Zhaoshan Project – HVDC CB

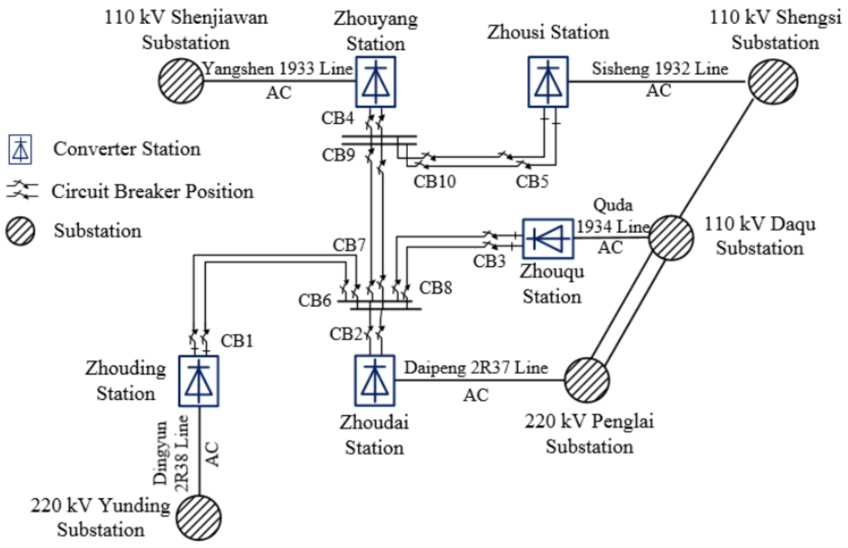
- **Main branch**: Conduct the load current
- **Commutation branch**: Interrupt the short-circuit fault current
- **Energy absorption branch**: Absorb the short-circuit current

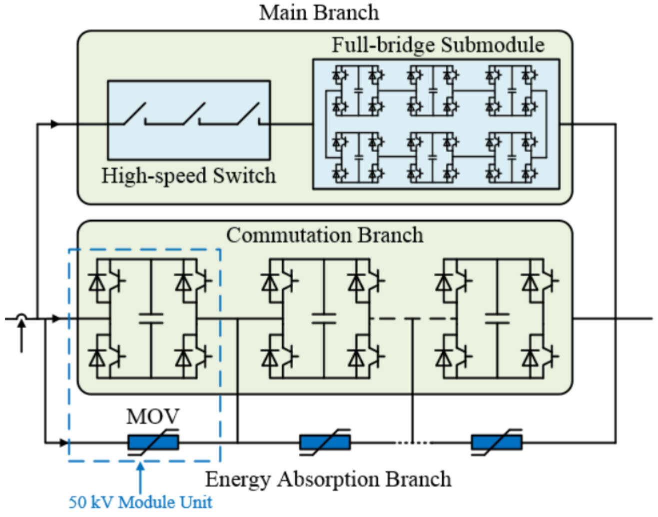
Steady-State Operating Condition

- Load current mainly passes through the main branch due to its lower conducting resistance.

DC Short-Circuit Fault

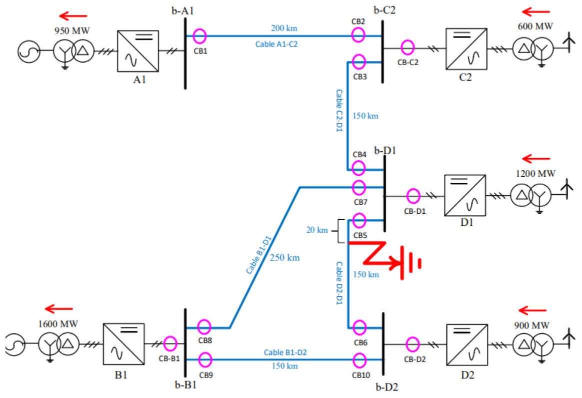
- IGBTs of full-bridge modules in the main branch will turn off, then the current is transferred to the commutation branch and drops rapidly to 0, and the high-speed switches are open. When the CB voltage reaches the protection threshold of the arrester, the current will be transferred to the energy absorption branch until it drops to 0, achieving current interruption and fault isolation.
- Interruption current of HVDC CB: 15kA 3ms.
- Withstand current of HVDC CB: 15kA, 5s; 40kA, 5ms
- Thermal Capacity: 8kA

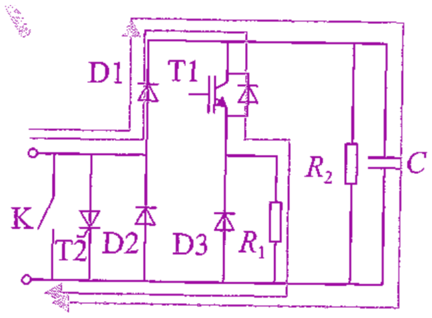


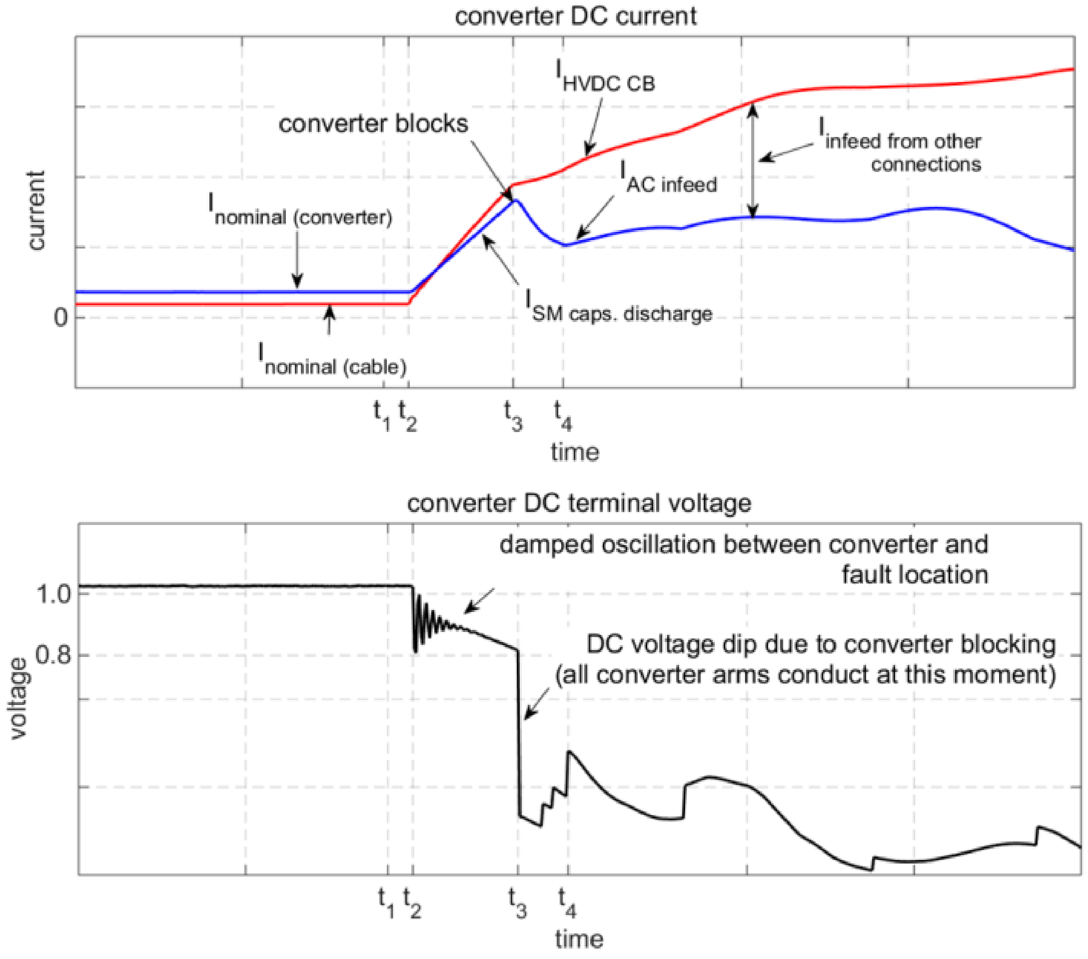


HVDC Fault Analysis – Equivalent Circuit in HB MMC

- **Fault Initiation**: A fault results in a voltage transient travelling along the cable and invokes the discharge of any charged capacitor (inc. the module capacitors, resulting in initial transients lasting only a few millisecond, before control action activated.
- **AC Source Contribution**: After the initial transient, AC sources start feeding the short circuit current, limited by AC side impedance and DC side resistance.
- **Current Limitation**: At HB MMC VSC terminal, the voltage transient discharge the submodule capacitors leading to a near-linear rise in current limited by converter arm reactor.
- **Converter Blocking**: To protect internal circuitry, an HB MMC VSC converter blocks, governed by protection algorithm combining arm and overall overcurrent and DC under-voltage. This turns the converter into an uncontrolled diode rectifier (=LCC).

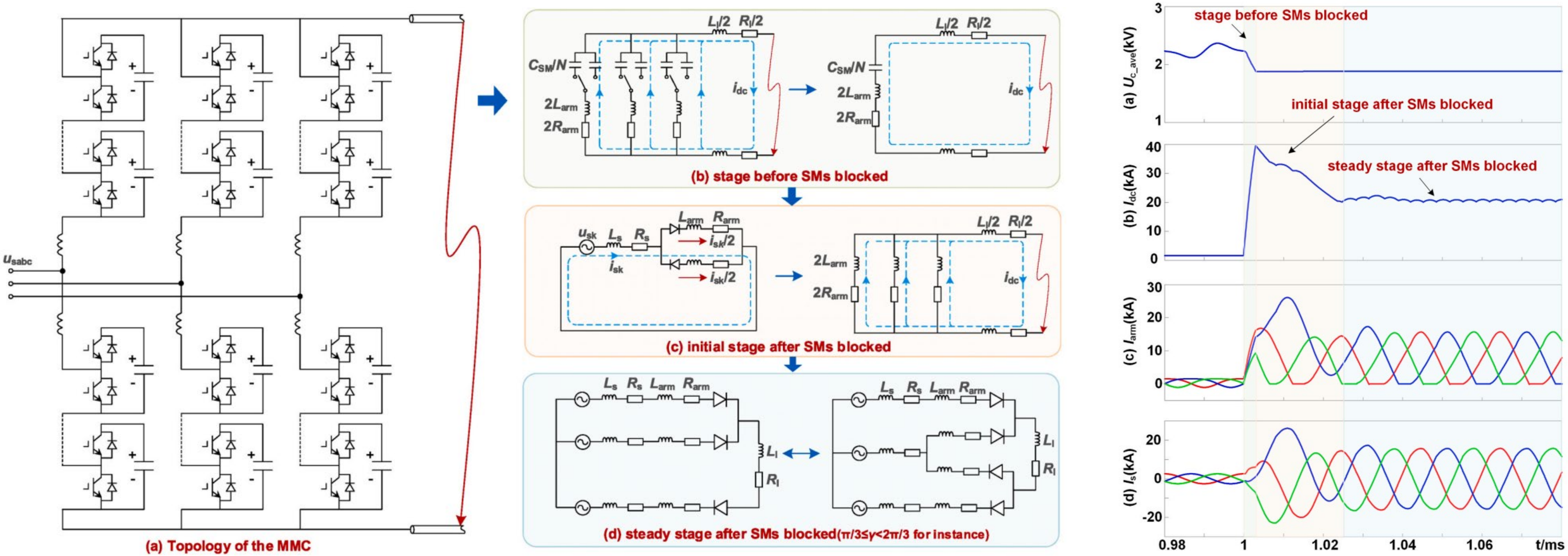






Fault Response of Converter and Fault Current at Location of HVDC CB

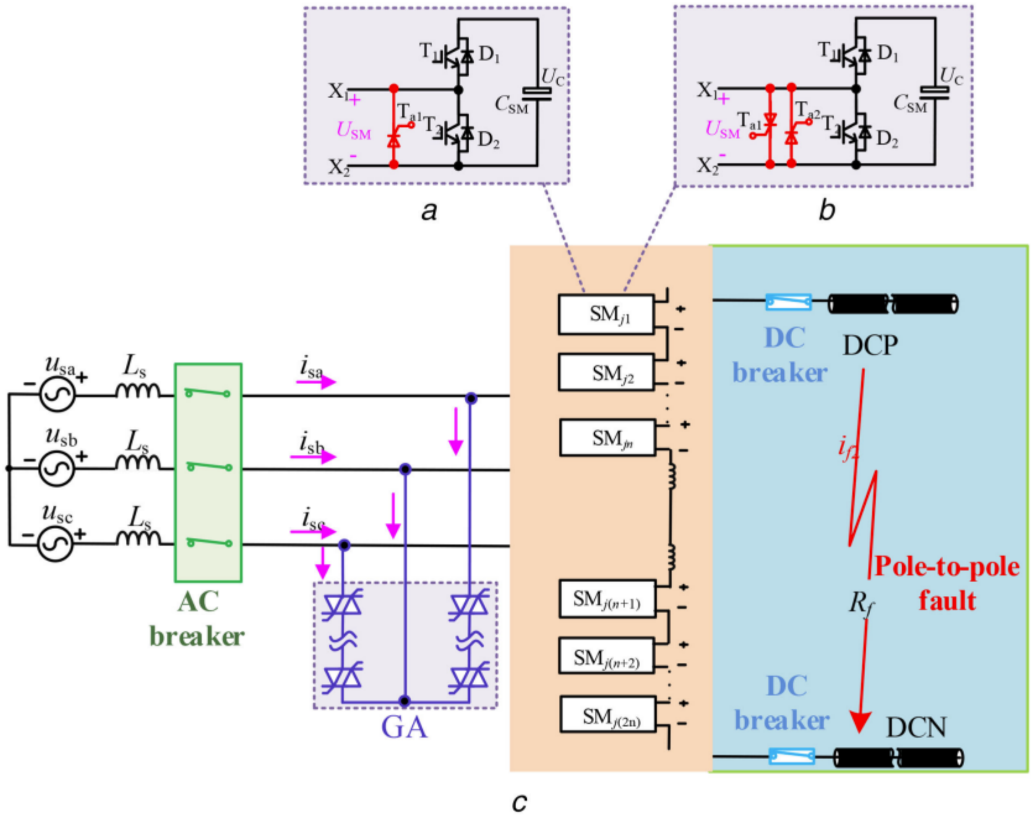
HVDC Fault Analysis – Equivalent Circuit in HB MMC

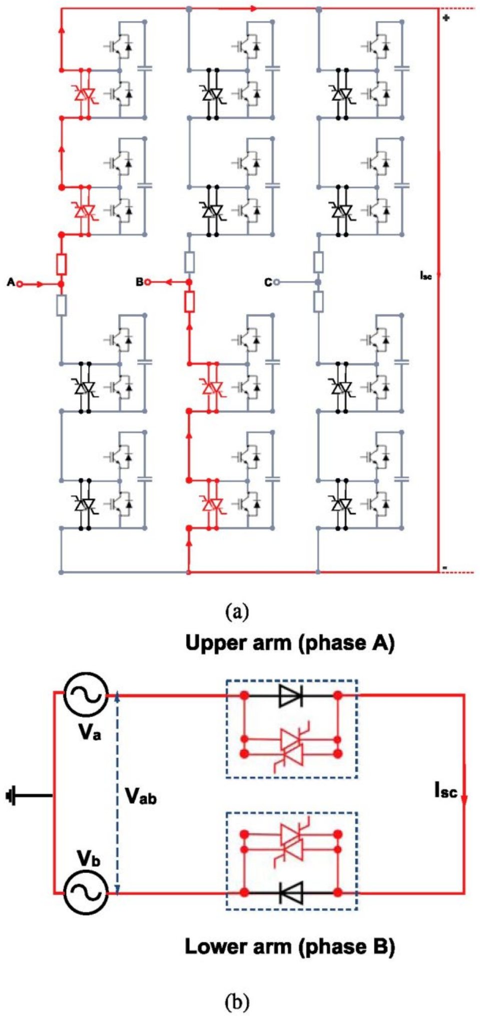


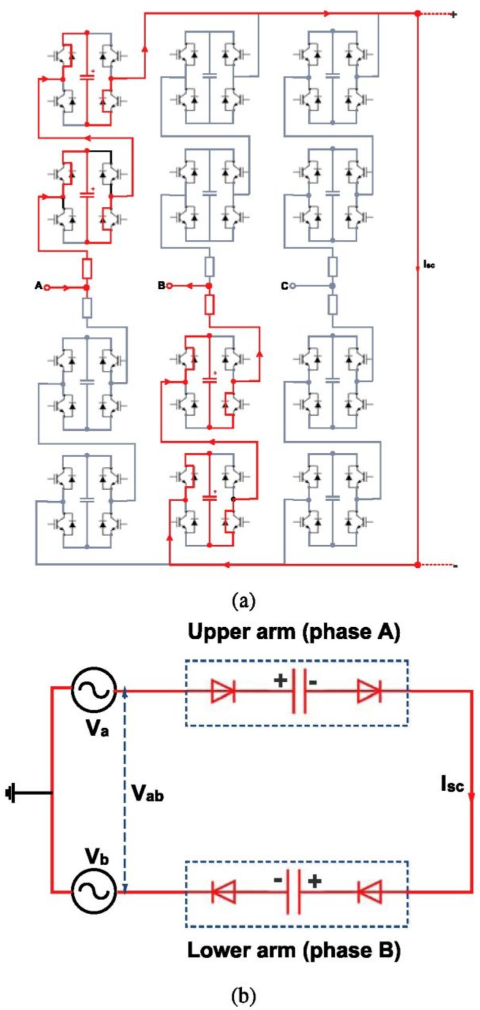
- **Before SMs Blocked**: Capacitor (C_{eq} = 6 x C_{SM} / N) discharges quickly to the fault point. The fault line is simplified into an RL model, and the MMC is modeled as RLC circuit. SM capacitors voltage decreased without control and monitoring, and current through the converter arm increases.
- **After SMs Blocked**: Diodes in blocked SMs conductor due to arm reactor's free-wheeling characteristics, resembling a three-phase fault from AC side. On the DC side, the fault current primarily consists of the arm reactor current, which decays in an RL first-order manner.

HVDC Fault Analysis – Equivalent Circuit in FB MMC

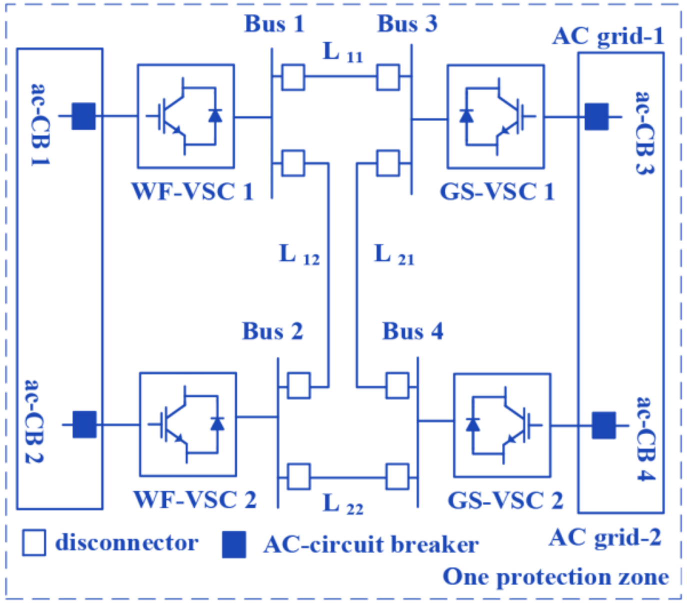
Note: DC Current Blocking is to block current from AC source continuous feeding.



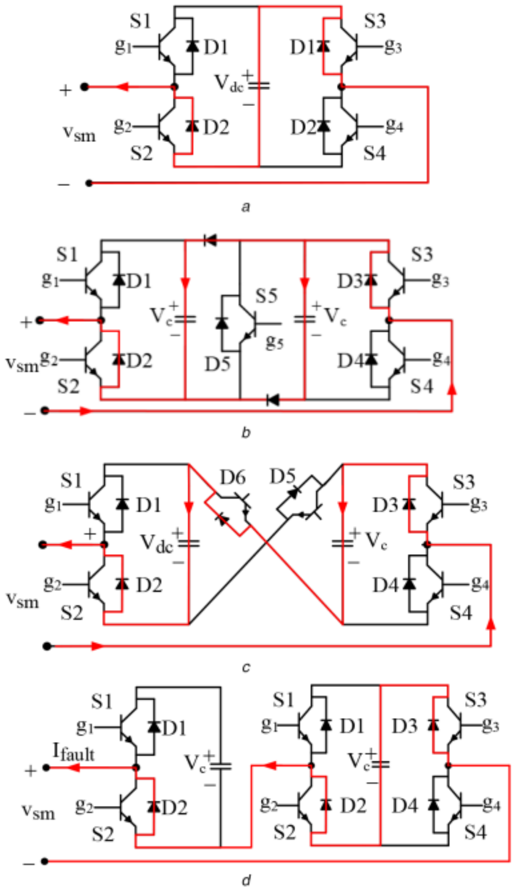




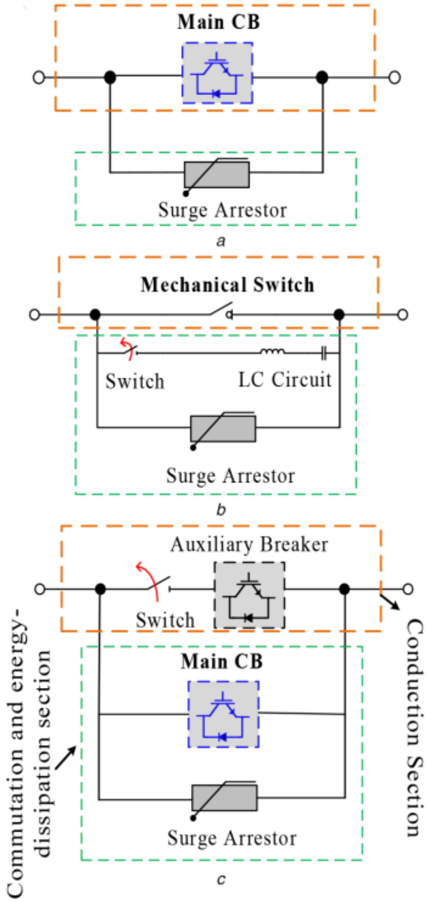
DC Fault Clearance



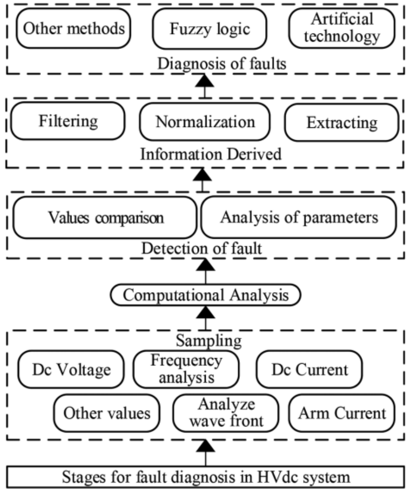
Fault Current Path with Blocked SM



Types of DC Circuit Breaker



Fault Diagnosis Process



DC Fault Clearance requires:

- Disconnection of Capacitor
- Isolation to DC side
- Isolation to AC side
- Action towards fault energy (dissipate by resistance / absorbed by source)

Detection Method:

- MMC arm current
- DC Overvoltage
- Capacitor Discharge
- Converter Output I
- Rate-of-Rise of DC
- DC Reactor Voltage
- Travelling Wave Theory
- Machine Learning

Fault Current Limiter (FCL) Application in HVDC

Requirement -

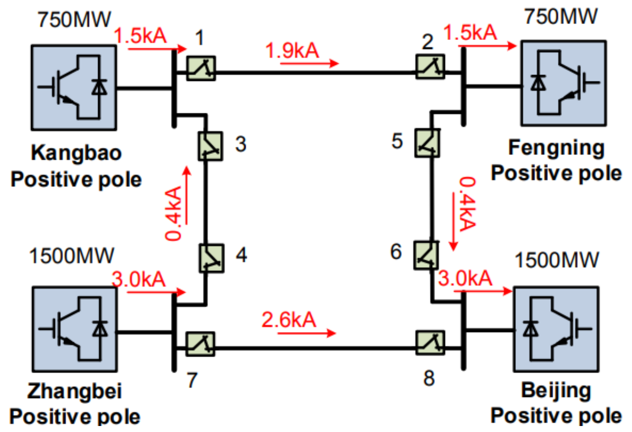
- **Act extremely fast** to limit the fault current growth.
- Ensure the fault current remains **below the tolerable threshold** of core equipment.
- Coordinate with **DCCB**, as it can **prolong the fault clearance time** due to restraining current decay.

Challenge -

- The **DC FCL should coordinate well with the DCCB** - it must limit fault current before the breaker operates and **remove the current-limiting component immediately afterward**.
- **Influence on power system stability**: Since power flow change in **VSC-HVDC** requires rapid variation in DC current, the **FCL could reduce transient response speed** and affect system stability.

Current Approaches in HVDC Grids

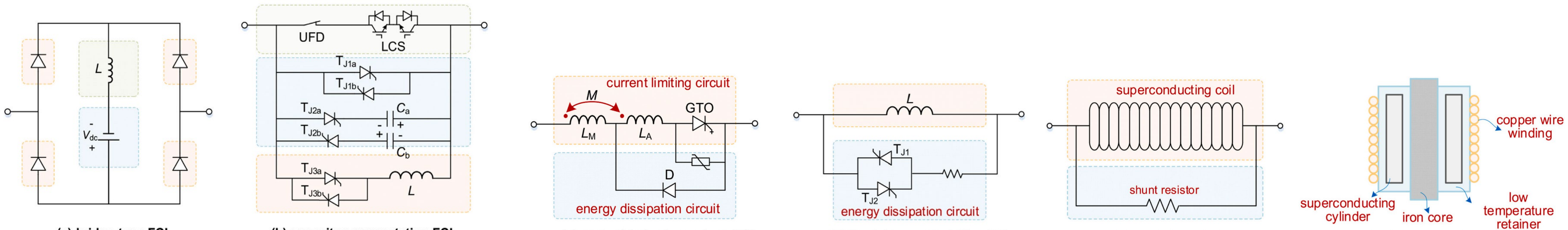
- The most common fault current-limiting strategy is the **direct installation of a DC reactor**, like the **150 mH reactors** used in the **Zhangbei DC grid, China**. While effective in limiting current rise, it has **drawbacks**, such as:
  - **Lack of coordination** with DCCB.
  - **Negative impact** on transient response and system stability.



THE MOST SERIOUS LINE TRANSIENT CURRENT AT DIFFERENT TIME

DC reactor	1ms/ kA	2ms/ kA	3ms/ kA	4ms/ kA	5ms/ kA	6ms/ kA	7ms/ kA	10ms/ kA
100 mH	6.6	10.5	13.6	14.7	18.0	19.5	19.7	22.4
150 mH	5.5	8.2	10.9	12.6	12.9	15.3	17.5	19.3
200 mH	4.8	7.0	9.1	11.1	12.9	13.0	13.7	18.0

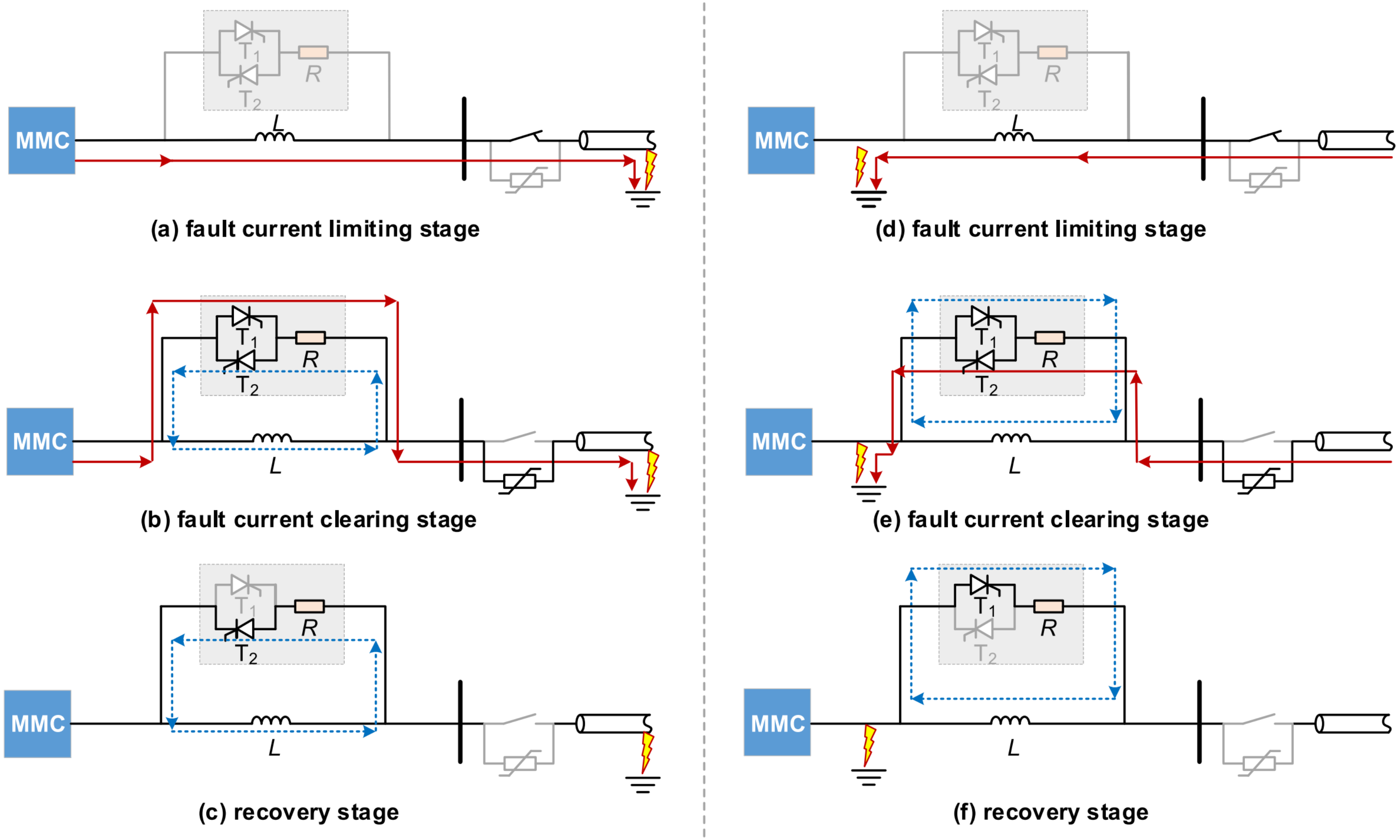
HVDC Fault Analysis – Fault Current Limiter (FCL)



(a) bridge-type FCL (b) capacitor commutation FCL (c) mutual-inductance-type FCL (d) thyristor commutation FCL (e) resistive SFCL (f) inductive SFCL

Type of FCL	Key Features	Pros	Cons
Power-Electronics-Based FCL	Uses **power electronic switches** (thyristors, IGBTs) for fast current limitation.	- Fast response (<1 ms) - Reduces MOV requirements in DCCBs	- Continuous conduction losses (bridge-type) - High cost (capacitor-based)
Bridge-Type FCL	Diode **H-bridge** + **DC reactor**; bypasses reactor during normal operation.	- Rapid activation - Limits peak fault current effectively	- Power loss in diodes - Requires pre-charged capacitors
Capacitor Commutation FCL	Uses **thyristors + capacitors** to divert fault current.	- Low conduction loss - Bidirectional operation	- Complex design - High cost (ultra-fast disconnectors)
Superconducting FCL (R-SFCL)	**Superconducting tape** quenches under fault, introducing high resistance.	- Zero impedance (normal operation) - Ultra-fast response	- Cooling system required - High material cost
Saturated Iron-Core SFCL	**DC-biased coils** control inductance during faults.	- No quenching needed - High reliability	- Large size - High DC bias power
Mutual Inductance FCL	Uses **coupled inductors** for adaptive current limiting.	- Low steady-state loss - Scalable design	- Limited fault current suppression

HVDC Fault Analysis – Fault Current Limiter (FCL)



Operation principle after forward fault

Operation principle after backward fault

Coordination between HVDC CB and FCL

Without Coordination:

- DCCB's MOV absorbs fault energies from both the fault line and DC reactors.
- Longer fault current clearing time: approximately **6ms**.
- Higher energy dissipation: **68.7 MJ**.

$$E_{\mathrm{MOV}} = \int_0^{t_{\mathrm{clear}}} u_{\mathrm{DCCB}} \cdot i_{\mathrm{dc}} \mathrm{d}t = \int_0^{t_{\mathrm{clear}}} U_{\mathrm{res}} \cdot \left(I_{\mathrm{dc\_max}} - \frac{I_{\mathrm{dc\_max}}}{t_{\mathrm{clear}}} t \right) i_{\mathrm{dc}} \mathrm{d}t$$
$$= U_{\mathrm{res}} \cdot I_{\mathrm{dc\_max}} \left[\left(t - \frac{t^2}{2t_{\mathrm{clear}}} \right) \Big|_{t=t_{\mathrm{clear}}} - \left(t - \frac{t^2}{2t_{\mathrm{clear}}} \right) \Big|_{t=0} \right]$$
$$= \frac{1}{2} U_{\mathrm{res}} \cdot I_{\mathrm{dc\_max}} \cdot t_{\mathrm{clear}} = \frac{1}{2} \cdot 1000\mathrm{kV} \cdot 22.9\mathrm{kA} \cdot 6\mathrm{ms} = 68.7\mathrm{MJ}$$

With Coordination:

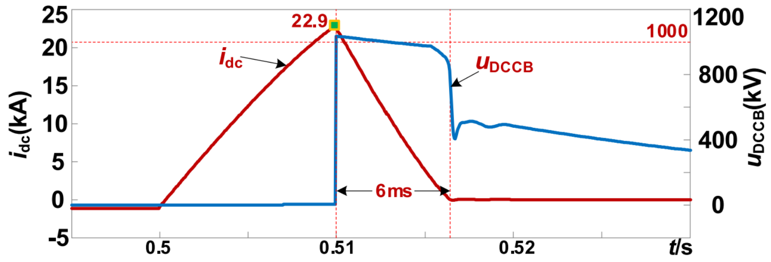
- FCL coordinates with DCCB by **bypassing the reactor after DCCB is tripped**.
- Reduced fault current clearing time: **1ms**.
- Lower energy dissipation: **11.45 MJ**.
- MOV-dissipating energy reduced by 85.5%.

Benefits:

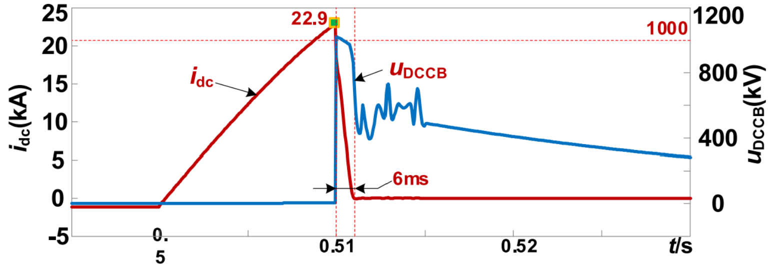
- Eases implementation and investment in high-voltage DCCBs.
- Calculation errors due to constant residual voltage assumption are within an acceptable range.

Superconducting Fault Current Limiters (SFCLs):

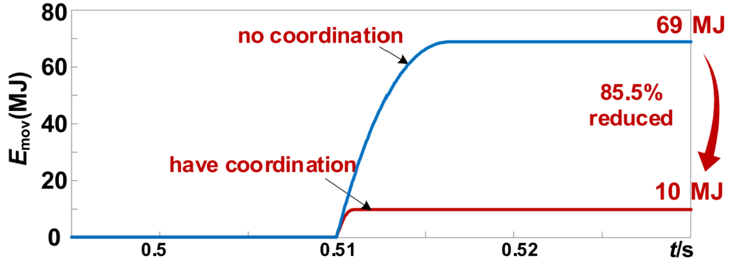
- Promising for VSC-HVDC applications.
- **Resistive SFCLs** show fast response times.
- Challenges: **cooling systems** and **recovery techniques**.



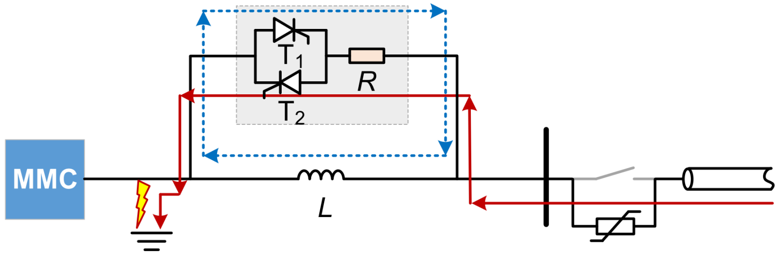
(a) no coordination between FCL and DCCB



(b) have coordination between FCL and DCCB



(c) absorbed energy of DCCB MOV

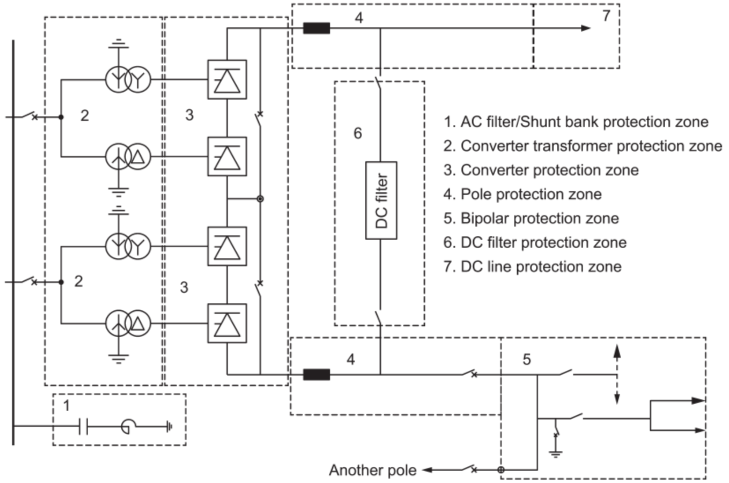
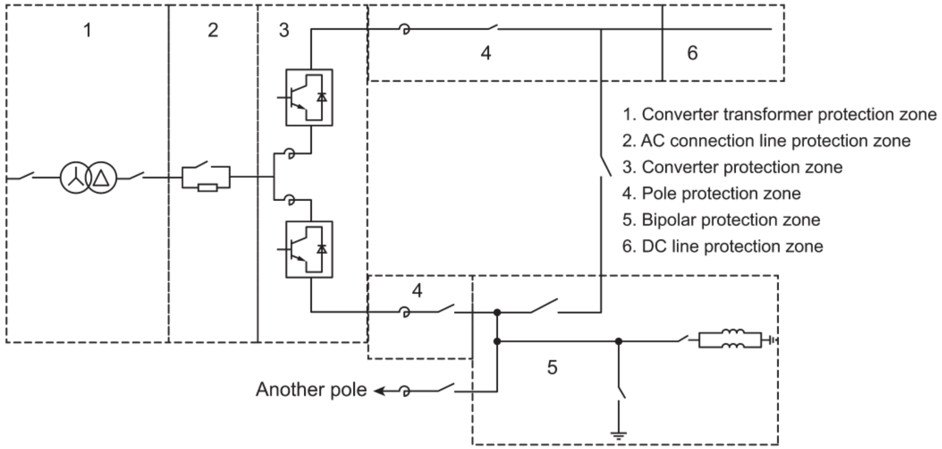
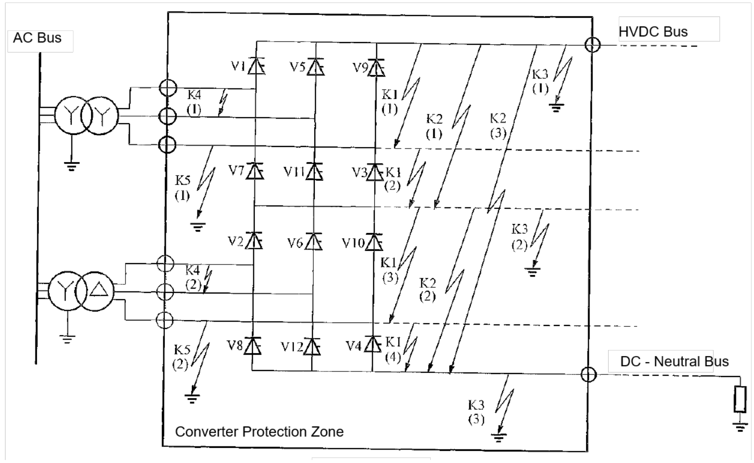
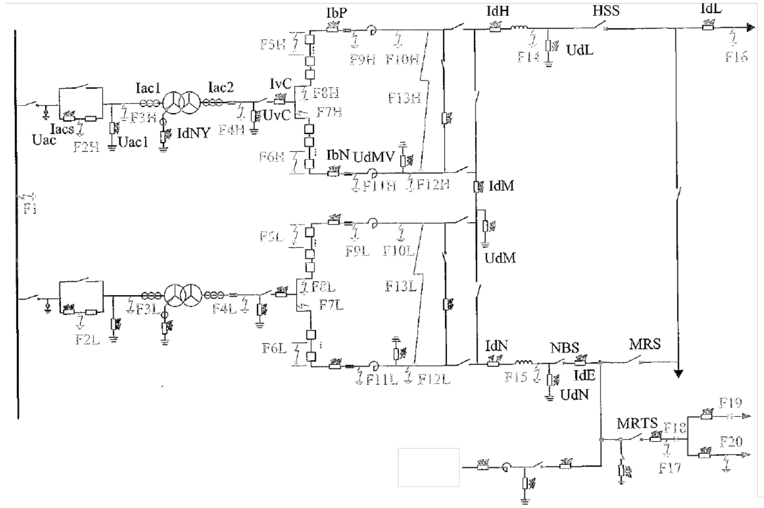


Control Action after Protection

Under protection actions upon fault or abnormal conditions, the following may result -

- ALARM
- REVERSE
- BYPASS 投旁對通
- BLOCK BYPASS
- FAULT RESTART
- CONTROL SYSTEM TRANSFER
- RUN BACK (REDUCE POWER OUTPUT / CURRENT OUTPUT)
- POLE BALANCE
- POLE ISOLATION
- ESOF (EMERGENCY SWITCH OFF)
- AC CB TRIP
- CLOSE NEUTRAL EARTHING
- BREAKER FAILURE INITIATION
- POLE BLOCK (NOT TO OPERATE DC SWITCH)

Zone Division and Fault Considered

	LCC – HVDC	MMC – HVDC
Zone Division -		
Possible Faults -		

Line Protection – LCC HVDC

1. **Fault Current Limitation**:
    - The initial fault current is limited by the surge impedance (around 2.0 p.u.).
    - The rectifier's current controller acts to reduce this current to the pre-fault level.
    - If the inverter's current control is active and power reversal is allowed, the steady-state fault current equals the current margin (approximately 0.1 p.u.), which is much lower than AC line fault currents.
2. **Fault Clearing**:
    - To clear the fault quickly, the inverter is prevented from operating as a rectifier, and the rectifier is switched to inverter mode by increasing the delay angle to its maximum (about 135^{o}), i.e. both sides in inverter mode.
    - Both converters help discharge stored energy in the DC circuit (L & C) back to the AC system, causing the DC line current and voltage to drop to zero, which aids in deionizing the arc path.
3. **Restart Process**:
    - After a brief delay (0.2 to 0.5 seconds), the line is automatically re-energized by restarting the converters, ramping up voltage and current.
    - If the fault persists, the protective action repeats. Typically, 3 restart attempts are made with increasing dead times.
    - If all attempts fail, the link is shut down until the fault is repaired. Alternatively, reduced voltage restart may be tried for insulation failures caused by pollution or bad weather.

Comparison with AC Systems:

- The process is analogous to fault clearing and auto-reclosing in AC lines but differs in that DC systems use converter control and protective relays instead of breakers.

Line Protection

Traveling Wave Protection

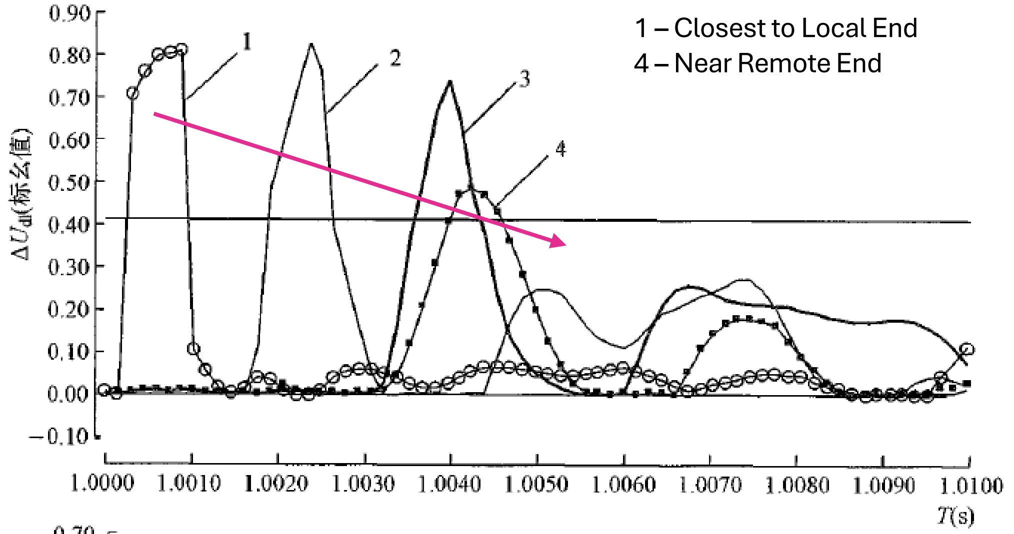
- **Principle**:
  Detects high-frequency voltage/current waves generated by faults, which propagate along the DC line at near-light speed.
- **Subtypes**:
  - **Voltage Traveling Wave Protection**:

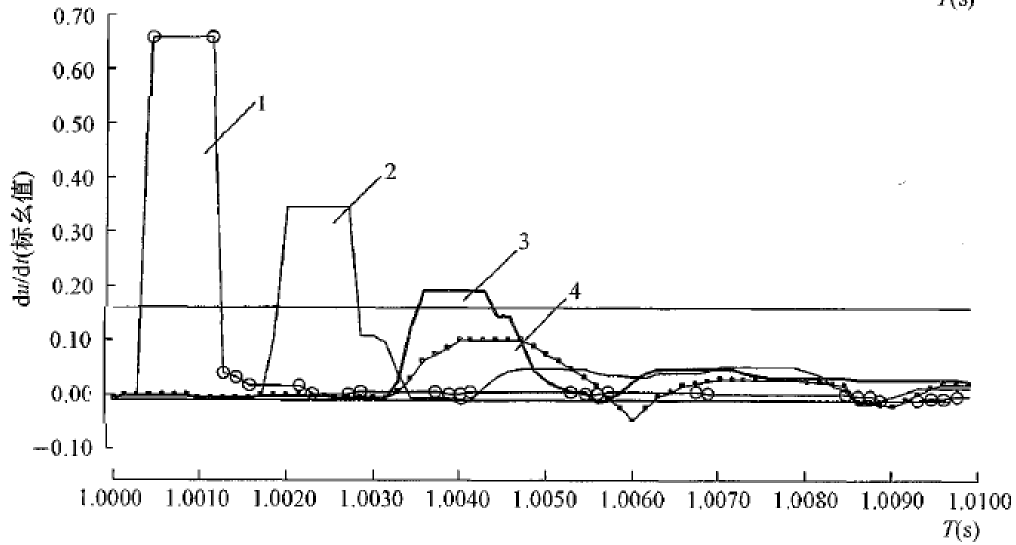
$$\begin{cases} \mathrm{d}u/\mathrm{d}t > \Delta_1 \\ \Delta u > \Delta_2 \\ \Delta_{i\_\mathrm{rec}} > \Delta_3 \, (\Delta_{i\_\mathrm{inv}} > \Delta_4) \end{cases}$$

    - **Thresholds**: $$\left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{\mathrm{set}} = K_{\mathrm{rel}} \cdot \left.\frac{\mathrm{d}u}{\mathrm{d}t}\right|_{\mathrm{out.\,max}}$$
    - **Action**: ESOF (no intentional delay).
  - **Current Traveling Wave Protection**:

$$\begin{cases} \mathrm{d}P_i/\mathrm{d}t > \Delta_1 \\ \Delta P_i > \Delta_2 \\ G_{\mathrm{wav}} > \Delta_3 \end{cases} \qquad \begin{cases} P_i = Z_\alpha i_{\mathrm{dl}i} - u_{\mathrm{dl}i} \\ G_{\mathrm{wav}} = Z_0 (i_{\mathrm{el}} + i_{\mathrm{N1}} + i_{\mathrm{N2}}) - (u_{\mathrm{dl1}} + u_{\mathrm{dl2}}) \end{cases}$$

    - **Application**: Discriminates fault direction and location.
- **Advantages**:
  - Ultra-fast response (<1 ms).
  - Covers the entire line length.
- **Limitations**:
  - Sensitivity drops for high-resistance faults (Rf > 50 Ω).

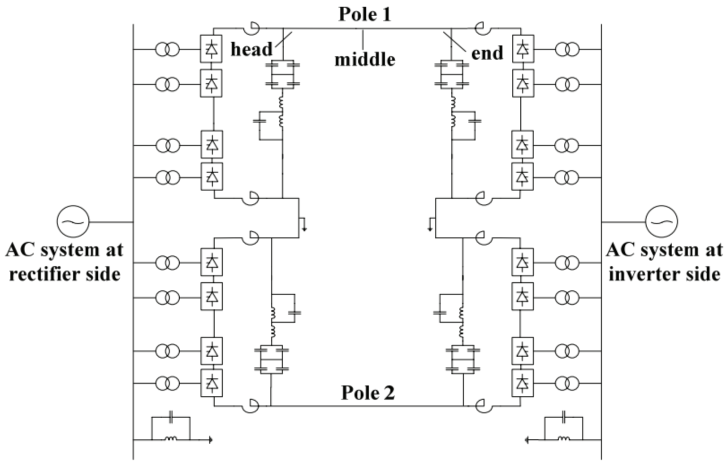


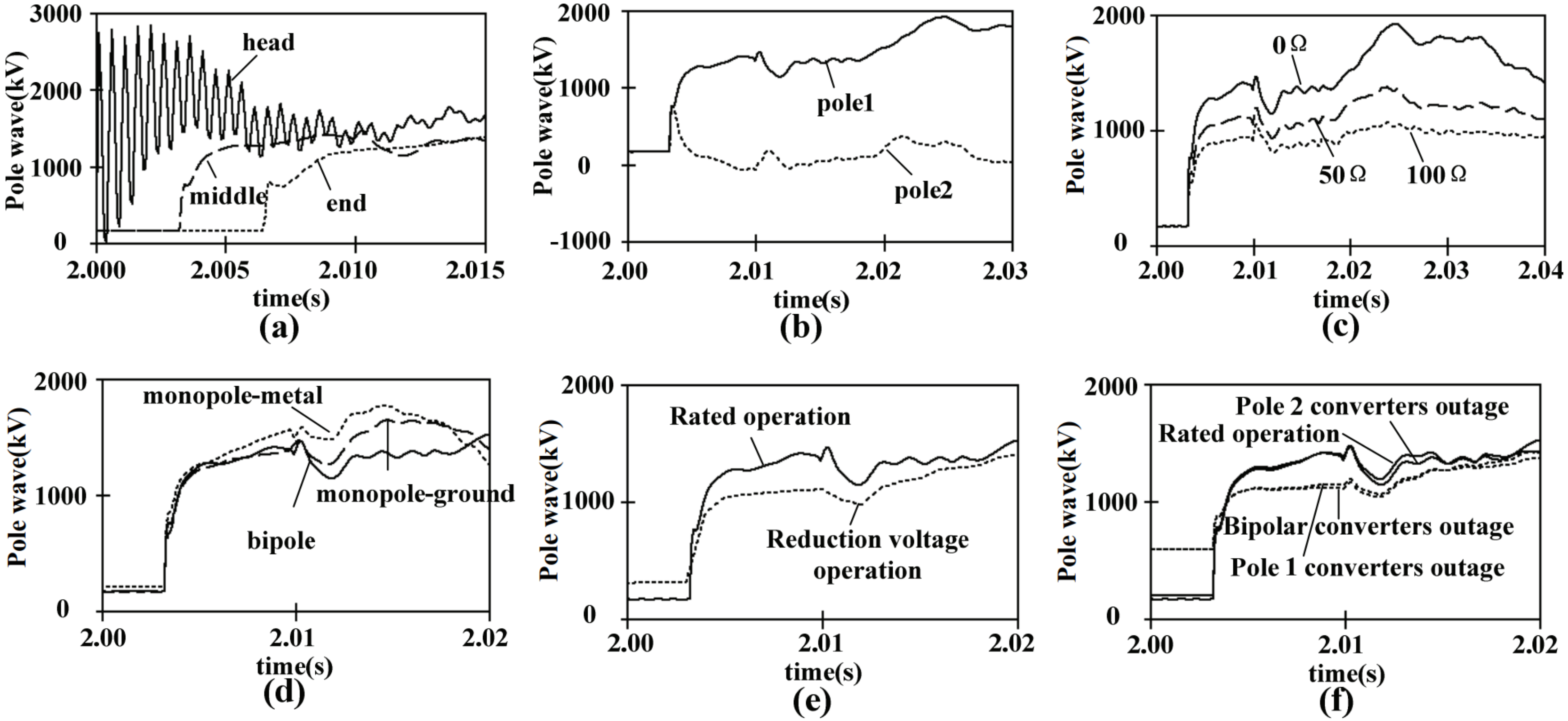


Characteristics of Pole Waves under Different Conditions

$$\left.\frac{du}{dt}\right|_{\text{set}} = \boxed{K_{\text{rel}}} \left.\frac{du}{dt}\right|_{\text{out. max}}$$

Challenges –
Determine sensitivity and reliability under nonlinearity of HVDC



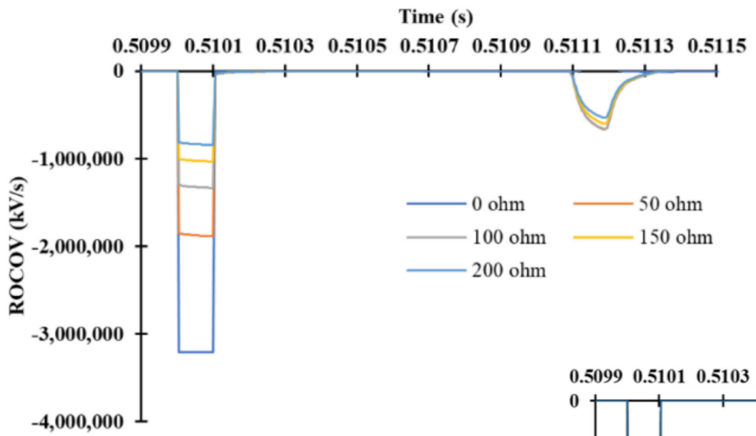


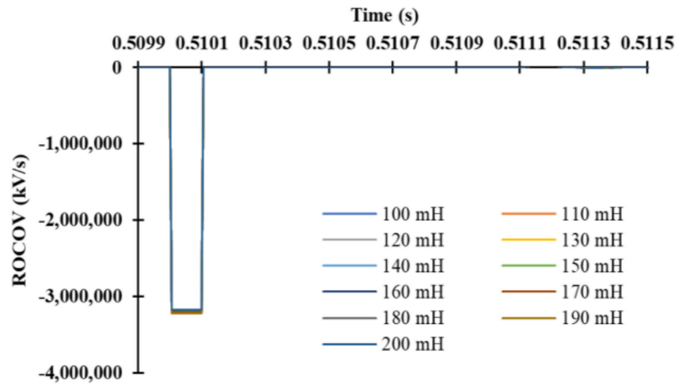
(a) Different Fault Distance
(b) Different Fault Pole
(c) Different Ground Resistance
(d) Same Operating Voltage
(e) Different Operating Voltage
(f) Different Converter Outages

Line Protection

(b) Differential Undervoltage Protection

- **Role**: Backup for traveling wave protection.
- **Principle**: Combines two criteria:
    - **Voltage Rate-of-Change** (du/dt<Δuset).
    - **Absolute Undervoltage** (ui<0.7–0.8 p.u).
- **Delay**: 10–50 ms (avoids nuisance trips during transients).
- **Action**: Triggers auto-reclosing or ESOF (Emergency System Off) for permanent faults.

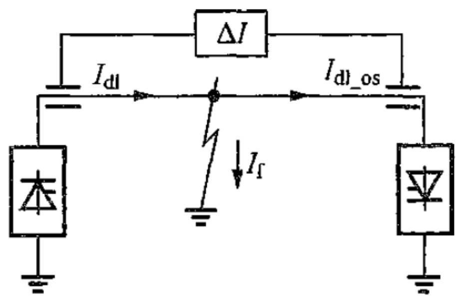


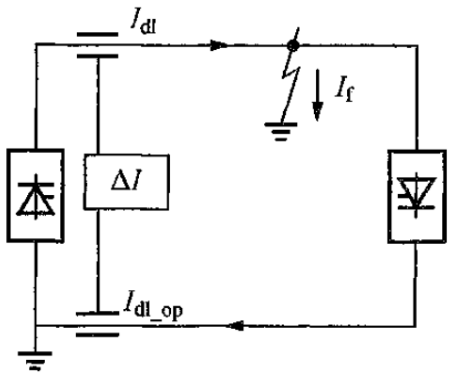


2. Backup Protections

(a) Line Differential Protection

- **Principle**: Compares currents at both ends of the DC line (rectifier vs. inverter): |Ia1−Ia2|>Δiset (ΔIset≈10–20% of rated current)
  Time Delay: 100–200 ms (longer than fault recovery time).
- **Purpose**: Detects high-resistance faults (Rf>100 Ω*Rf*>100 Ω) missed by traveling wave protection.

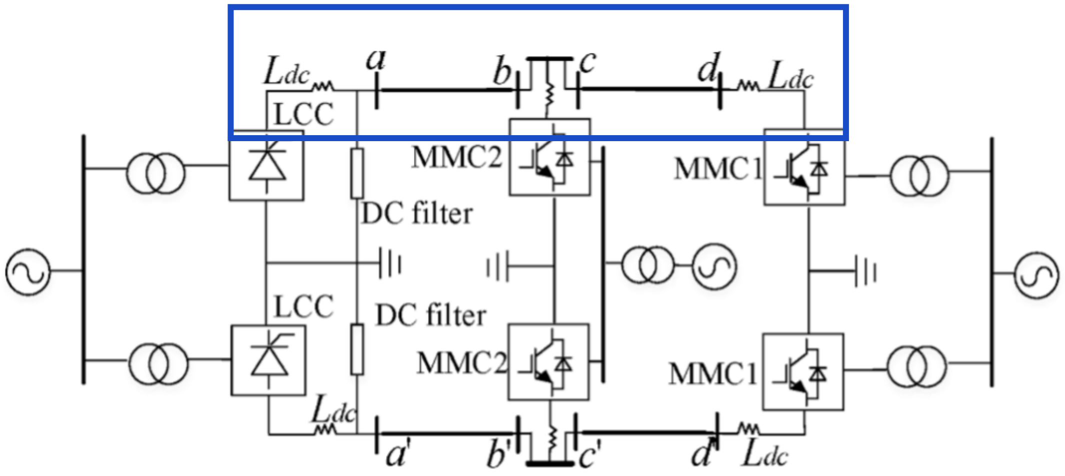


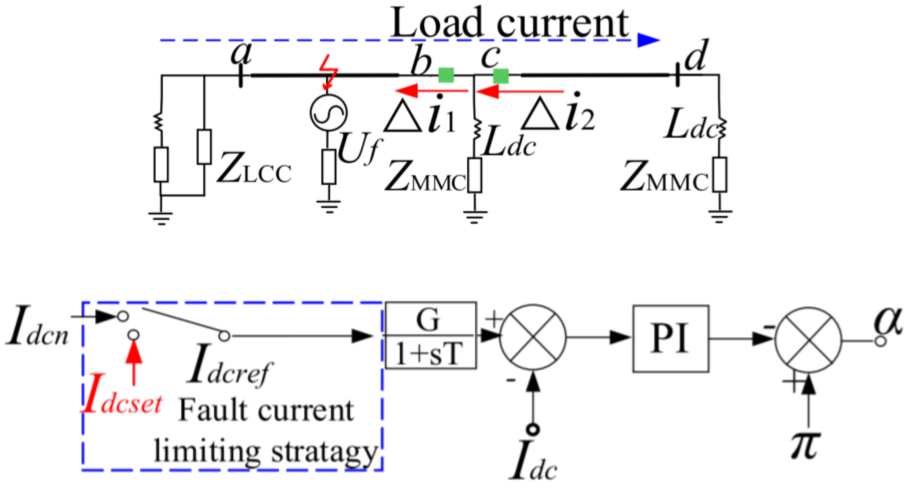


(b) Pole Bus Differential

- **Role**: Compares currents at different poles (+DC to -DC)
- **Action**: Direct ESOF (no auto-reclose).

HVDC Protection – Control Affecting Fault Current and Stability





With **Full Bridge** Submodules, it provides **negative voltage level** output capacity, the sum of differential equation of 3-phase bridge arm current can be obtained by

$$-R_{arm}i_{dc} - L_{arm}\frac{di_{dc}}{dt} + \frac{3}{2}U_{dref} = \frac{3u_{dc}}{2}$$

When fault impedance exists in fault branch, the equation of DC side fault is as follows.

$$R_{dc}i_{dc} + R_f i_{dc} + L_{dc}\frac{di_{dc}}{dt} = \frac{u_{dc}}{2}$$

Fault Current Limiting by Gate Shutting

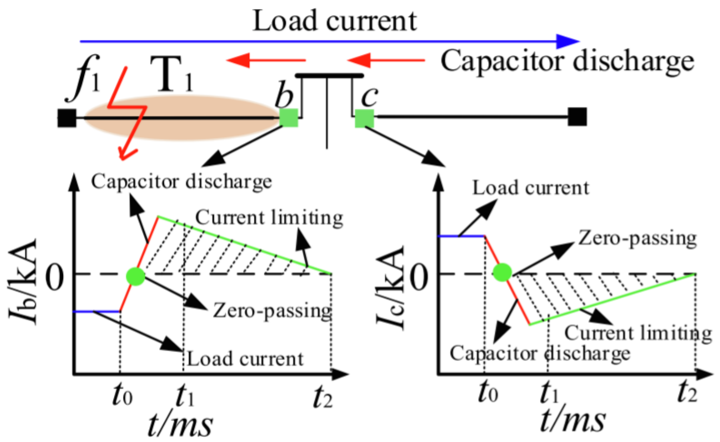
Combining the equations, the DC side current $i_{dc}(t)$ of the DC side port of MMC can be expressed as

$$i_{dc}(t) = \left[i_{dc(2)} - i_1(t_1)\right]e^{-\frac{(R_{arm}+3R_{dc}+3R_f)}{L_{arm}+3L_{dc}}(t-t_1)} + i_1$$

$$i_1 = \frac{3}{2}\frac{U_{dcref}}{R_{arm}+3R_{dc}+3R_f}$$

HVDC Protection – Control Affecting Fault Current and Stability

Fault at T_1

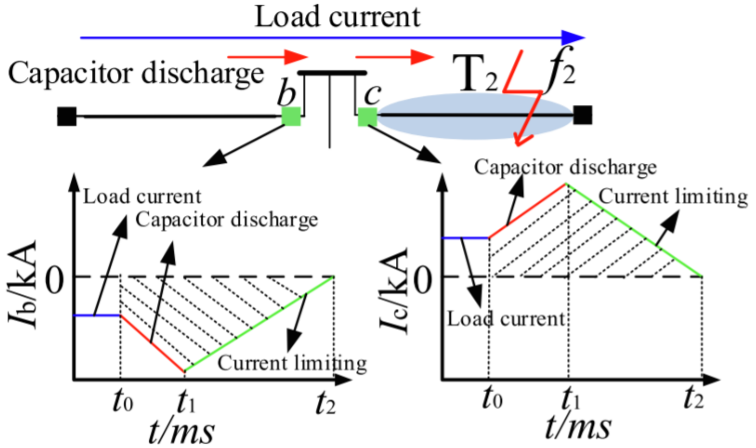


Due to **load current** at initial moment of fault, fault current cannot be changed suddenly with inductance. When load current is ignored, *b* should have current in greater **magnitude** than *c* with **opposite polarity**. Hence

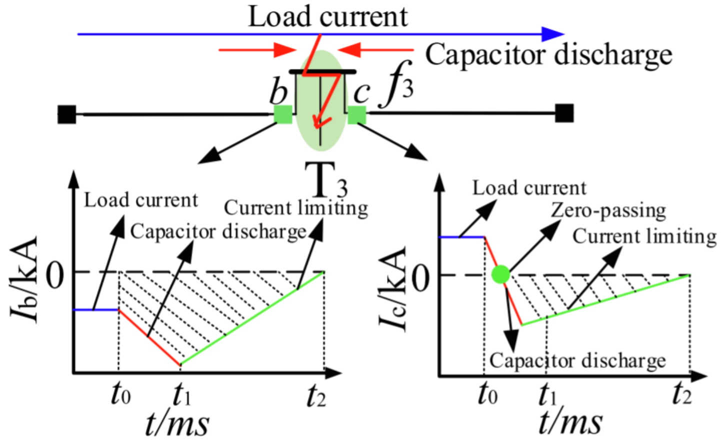
$$\sum_{i=1}^{n} \frac{I_b(i)}{I_c(i)} < 0, \qquad \sum_{i=1}^{n} I_b(i) > 0, \sum_{i=1}^{n} I_c(i) < 0$$

Where n is the sampling point.

Fault at T_2



Fault at T_3



Similarly for T_2,

$$\sum_{i=1}^{n} \frac{I_b(i)}{I_c(i)} < 0$$

$$\boxed{\sum_{i=1}^{n} I_b(i) < 0, \sum_{i=1}^{n} I_c(i) > 0}$$

Conclusion

HVDC with multiple sections allow **fault sectionalizing** with current sum polarity.

HVDC Protection – Control Affecting Fault Current and Stability

Trigger Criterion

$$\frac{1}{m}\sum_{k=1}^{m}|I(k)| > I_{set} \quad \text{or} \quad \frac{1}{m}\sum_{k=1}^{m}|U(k)| < U_{set}$$

Consider the coupling effect of bipolar, it is necessary to increase the criterion of **fault pole selection**.

Positive PTG:

$$\frac{\frac{1}{m}\sum_{k=1}^{m}|I_p(k)|}{\frac{1}{m}\sum_{k=1}^{m}|I_N(k)|} > k_{set1}$$

Negative PTG:

$$\frac{\frac{1}{m}\sum_{k=1}^{m}|I_p(k)|}{\frac{1}{m}\sum_{k=1}^{m}|I_N(k)|} < k_{set2}$$

PTP:

$$k_{set2} < \frac{\frac{1}{m}\sum_{k=1}^{m}|I_p(k)|}{\frac{1}{m}\sum_{k=1}^{m}|I_N(k)|} < k_{set1}$$

where k_{set1} and k_{set2} are set to 1.2 and 0.8 respectively.

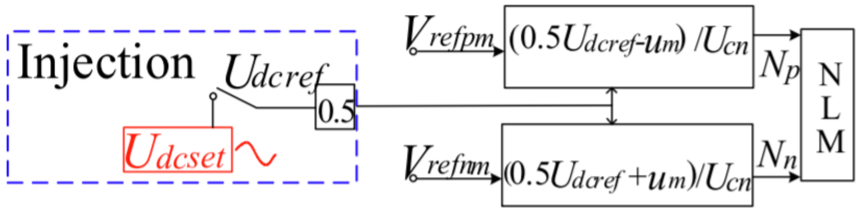
Protection Criterion

When a positive PTG fault or PTP fault occurs on DC system, combining with the result of fault pole selection, the polarity characteristics of b and c helps **identify faulty area**.

$$T_1: \begin{cases} \sum_{i=1}^{n}\frac{I_b(i)}{I_c(i)} < 0 \\ \sum_{i=1}^{n} I_b(i) > 0\,, \sum_{i=1}^{n} I_c(i) < 0 \end{cases} \qquad T_2: \begin{cases} \sum_{i=1}^{n}\frac{I_b(i)}{I_c(i)} < 0 \\ \sum_{i=1}^{n} I_b(i) < 0\,, \sum_{i=1}^{n} I_c(i) > 0 \end{cases}$$

Signal Injection Process

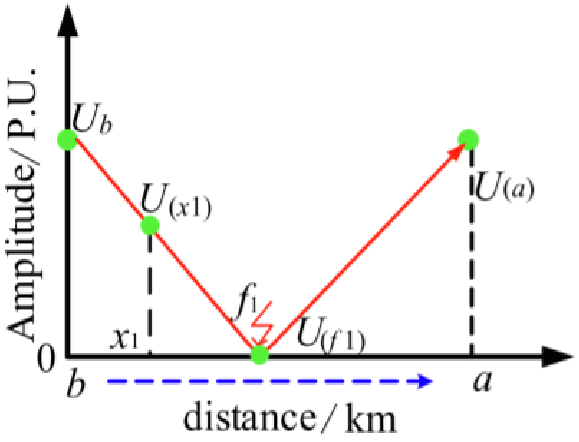
It is to aid converter blocking with signal injection.

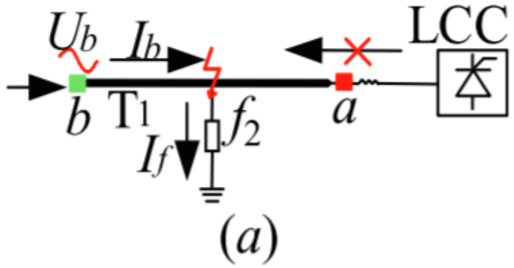


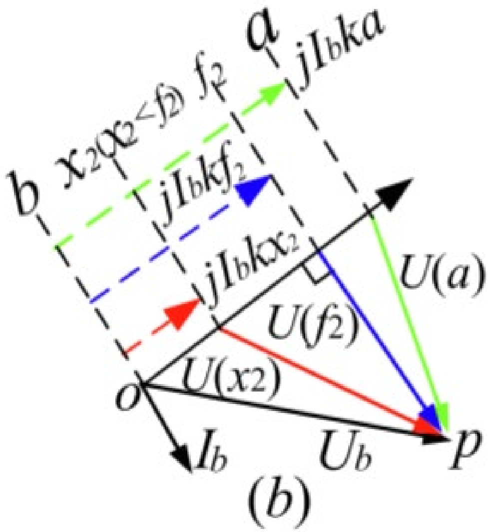
HVDC Protection – Control Affecting Fault Current and Stability

Voltage Based Fault Location

Considering the amplitude of signal injection, fault location can be executed.





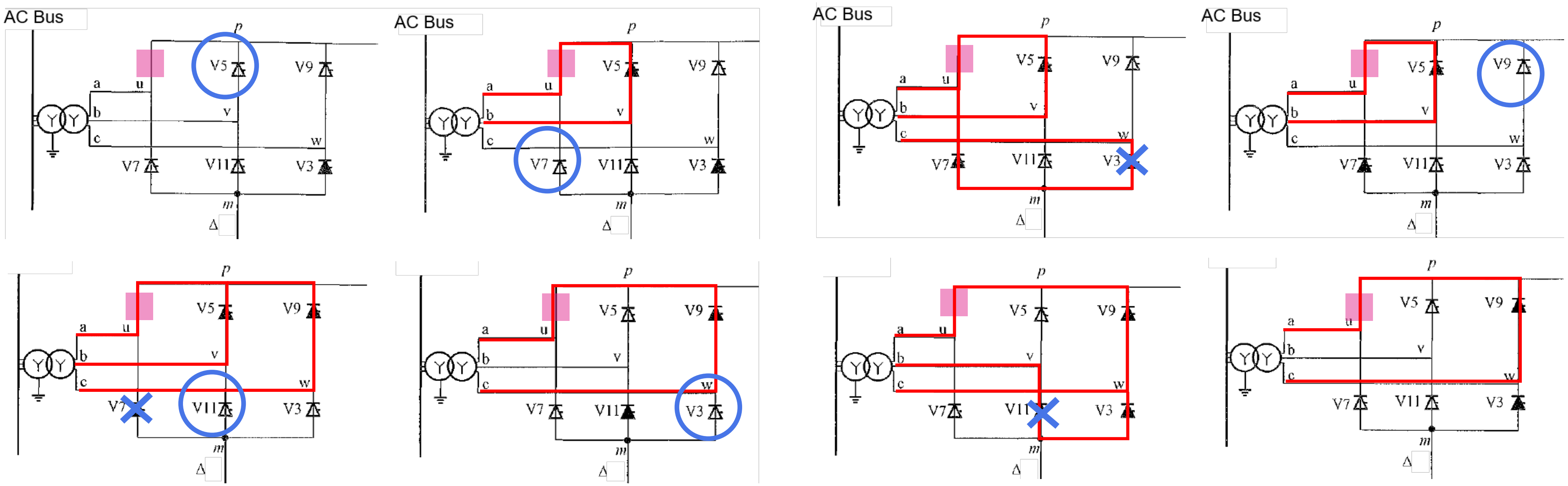


Impedance Based Fault Location

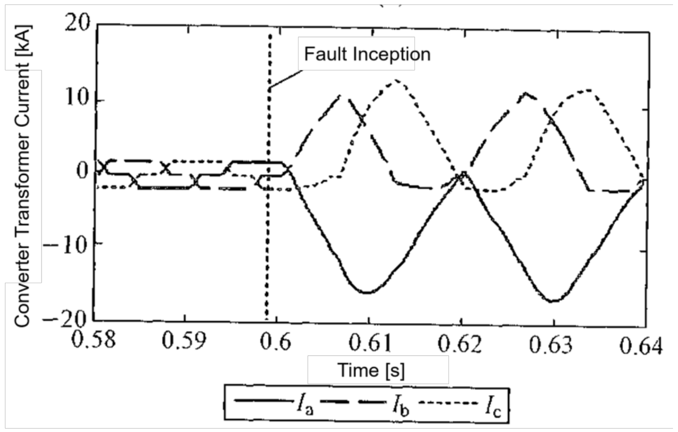
When x_2 is shorter than the fault distance f_2, the amplitude of voltage $U(x_2)$ can be expressed as

$$|U(x_2)| = |U_b - jI_bkx_2|$$

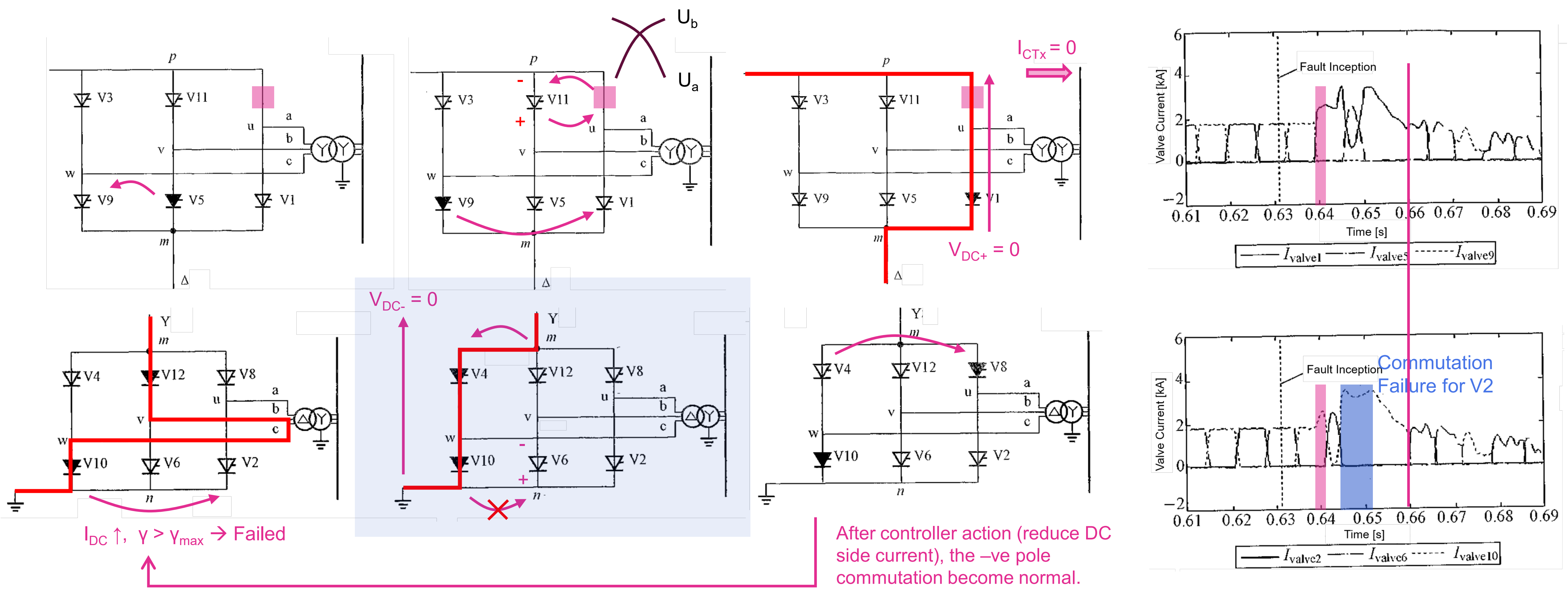
LCC – Converter Fault: Rectifier Valve Failure



- Rectifier end short circuit fault has a periodic (and continuous) waveform according to switching pattern.
- The development of short circuit fault affects the current through converter transformer.
- 5 – 6 times and 8 – 9 times of rated current goes through the healthy thyristor at upper arm and faulted thyristor respectively.
- Converter transformer has a continuous and periodic current with magnitude 8 – 9 times of nominal through, and the positive half and negative half cycle is asymmetric with DC component, possibly damaging the transformer.
- DC side has 50Hz AC component.
- The fault in AC side is a switched fault (L-L and 3 phase fault), and the fault is dependent on the firing angle α.

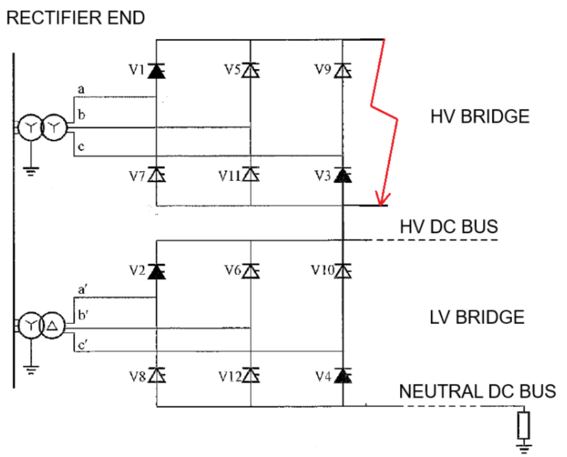


LCC – Converter Fault Inverter Valve Failure

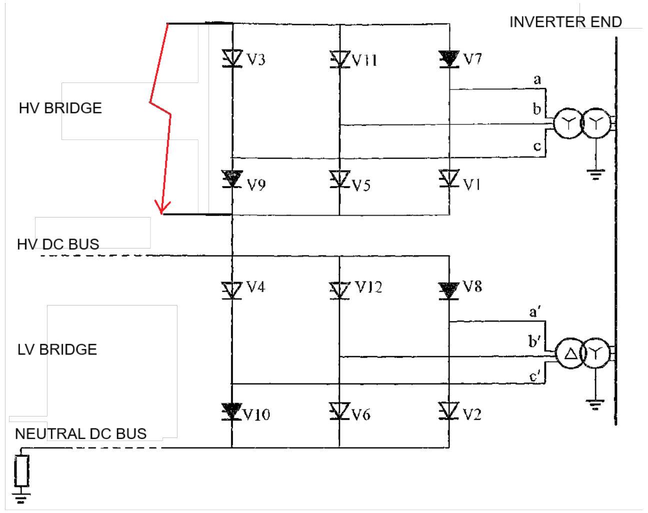


- Commutation failure at HV pole(V7) can lead to commutation failure of a healthy valve at LV pole(V2) due to increased DC side current with extinction angle γ larger than its maximum. DC side current under control action is increased, but to a limited value.
- Under current limiting action at LV pole, fault current at negative pole will decrease to value lower than nominal value, and the commutation failure will be self healed.
- The phase reversal action (倒换相) from a healthy valve to a fault valve occurs periodically and it introduces harmonics to DC side.
- Asymmetrical current (with DC offset) flows through converter transformers at inverter ends and it leads to overheat and noise in the transformers.

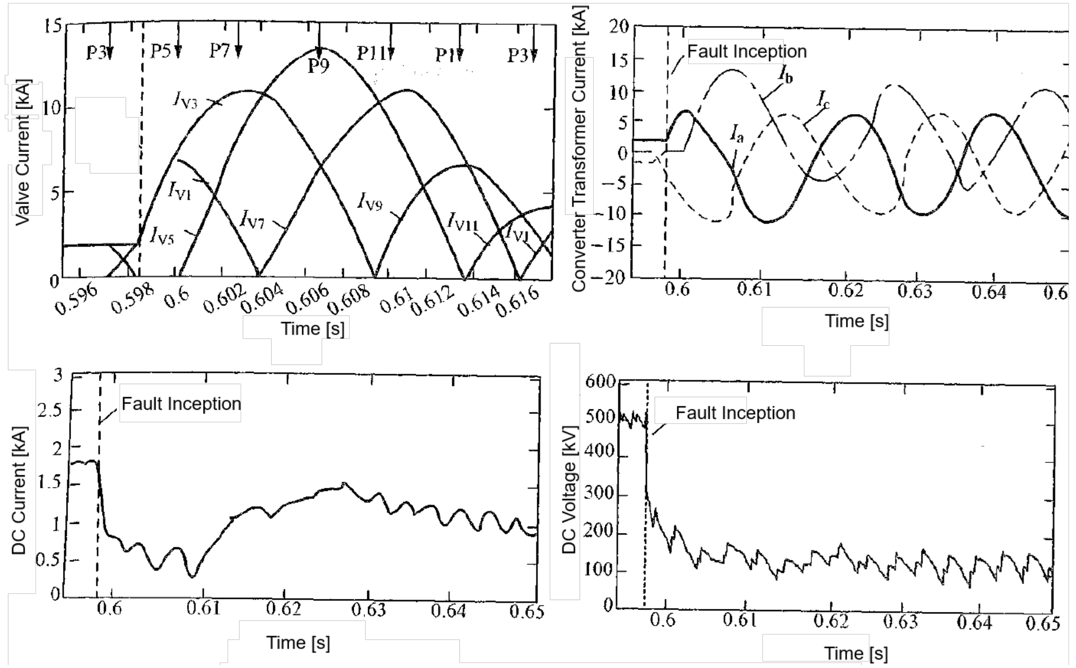
LCC – Short Circuit (P-N) at Rectifier End vs Inverter End



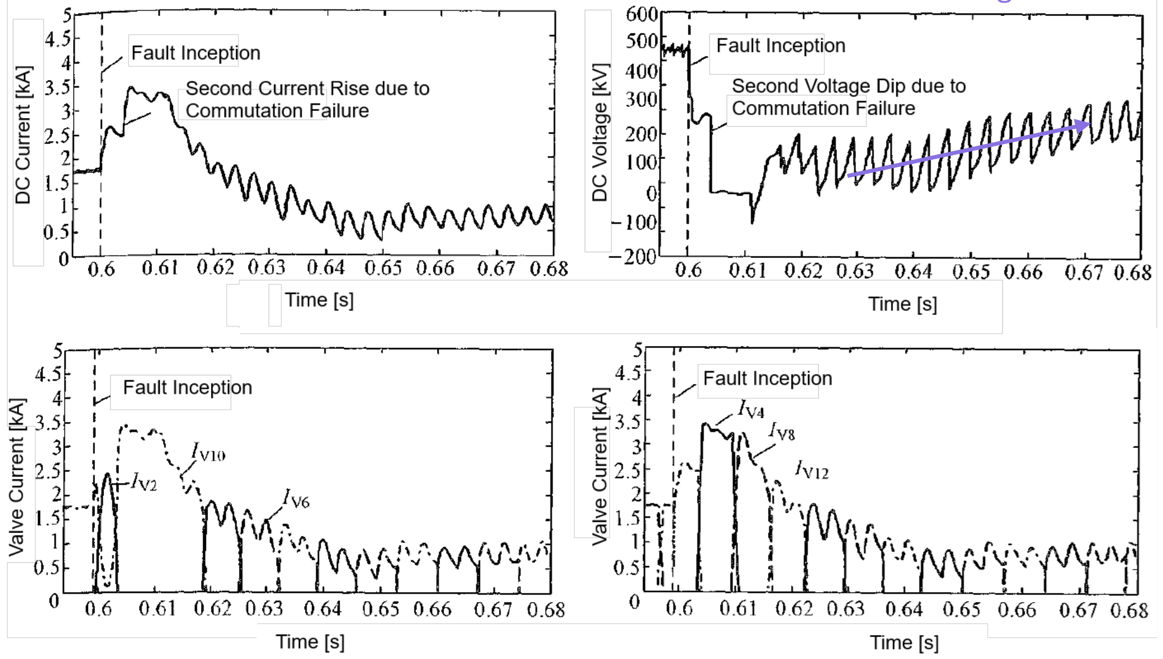
Switched Fault
(a/c and a/b/c)



Switched Fault
(Circulating Path at
DC side vs AC side)



Limited by
Smoothening Reactor



Commutation Failure

Requirements for Successful Commutation

For successful commutation in **Line-Commutated Converter (LCC) HVDC systems**, the following conditions must be met:

1. **Minimum Negative Voltage-Time Area**:
   - The thyristor valve must experience sufficient reverse voltage for a minimum duration (**γ, extinction angle**) to allow deionization of charge carriers (recombination).
   - Expressed as:

$$\gamma = \arccos\left(\frac{\sqrt{2}kI_dX_c}{U} + \cos\beta\right) - \phi$$

2. **Adequate AC Voltage**:
   - The inverter-side AC voltage (U) must be stable. A drop in voltage reduces γ, risking commutation failure.
3. **Controlled DC Current (Id)**:
   - Excessive Id increases the commutation overlap angle, reducing γ.
4. **Precise Trigger Timing**:
   - Firing pulses must be accurately synchronized to the AC voltage waveform.

Cause of Commutation Failure

External Factors (AC System-Related)

1. **AC Voltage Sag/Dip**: Short circuits, transformer faults, or grid instability near the inverter. (γ∝U)
2. **Asymmetric Faults**: Single-phase/ground faults distort voltage waveforms, introducing phase shifts (ϕ).
3. **Weak AC Grid**: Low short-circuit ratio (SCR) or high grid impedance (Xc).

Internal Factors (DC System-Related)

1. **High DC Current (Id)**:Overload or control system errors.Increases overlap angle, leaving insufficient time for deionization (γ∝1/Id)
2. **Firing Angle (β) Issues**: Incorrect control settings or delayed pulses reduces γ (Eq. 1).
3. **Thyristor Valve Faults**: Ageing, overheating, or manufacturing defects. Failure to block forward voltage after commutation.
4. **Harmonics/Resonances**: Poor filter design or grid interactions.

Multi-Infeed Interactions (Specific to Guangdong Grid)

- **Coupling Effect**: Proximity of multiple LCC-HVDC inverters in the Pearl River Delta exacerbates commutation failures. A fault affecting one inverter may propagate to others via shared AC bus voltage dips.

Commutation Failure in Multi-Terminal LCC – HVDC System

- Commutation Failure is failure of current transfer from one valve to next due to the reduction in commutation voltage of converter after any significant fault or disturbance.

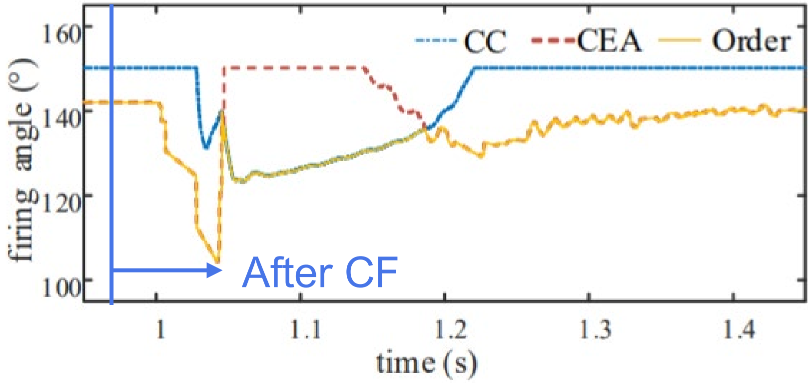
Method to Reduce Commutation Failure:

1. Installation of Voltage Support Equipment,
   e.g. STATCOM, Synchronous Condenser, Capacitor Bank.
2. Increase of AC System Strength
3. Stabilize Input Voltage
   e.g. with Capacitor Commutator Converter (CCC)
4. Reduce Fault Impact
   e.g. with Superconducting Fault Current Limiter
5. Advanced Control in Firing Angle
   e.g. Reduce Firing Angle

Yet, it often requires accurate commutation failure detection or even CF prediction.

Detection of Commutation Failure

1. Calculation of Maximum Acceptable Voltage Drop
2. Electromagnetic Transient (EMT) Simulation
3. Examination of Valve Conduction Status
4. Comparing Valve Current with DC Current
5. Estimate AC Current of All Phase in Transformer
6. Inspection of DC current of Converter
7. Observation of Extinction Angle



Valve Failure

1. Arc-Through (Arc-Back)

An **unintentional thyristor conduction** where a valve fires prematurely or fails to block forward voltage, causing current to flow outside its designated conduction period.

Cause:

- Insufficient reverse voltage (due to low AC voltage or commutation failure).
- High **dv/dt** (rate of voltage rise) triggering spurious firing.
- Thyristor gate pulse leakage or insulation failure.

Consequences:

- Short-circuit current flows through the valve, stressing the converter.
- Can lead to **commutation failure** or DC current disruption.

2. Misfire

A **failure to fire** a thyristor valve when it should conduct, typically due to missing or inadequate gate pulses.

Cause:

- Faulty gate pulse generation or delivery (e.g., control system error, damaged thyristor).
- Low AC voltage reducing thyristor forward bias.

Consequences:

- Disrupted current path in the converter bridge.
- Causes **asymmetric current waveforms**, leading to:
  - DC voltage/current ripple.
  - Increased stress on other valves.
  - Potential commutation failure in subsequent cycles.

3. Current Extinction

The natural **cessation of current** in a thyristor valve due to the **reversal of AC voltage** (end of conduction interval).

Process:

- Thyristors require zero current to turn off (natural commutation).
- Current extinction occurs when:
  - The AC voltage reverses polarity.
  - The commutation margin angle (γ) is sufficient to ensure thyristor recovery.

Consequences of Poor Extinction:

- If current fails to extinguish (e.g., due to low γ or AC voltage dip), commutation failure occurs.
- Results in DC current collapse or distortion.

Protection for Converter Protection Zone in LCC - HVDC

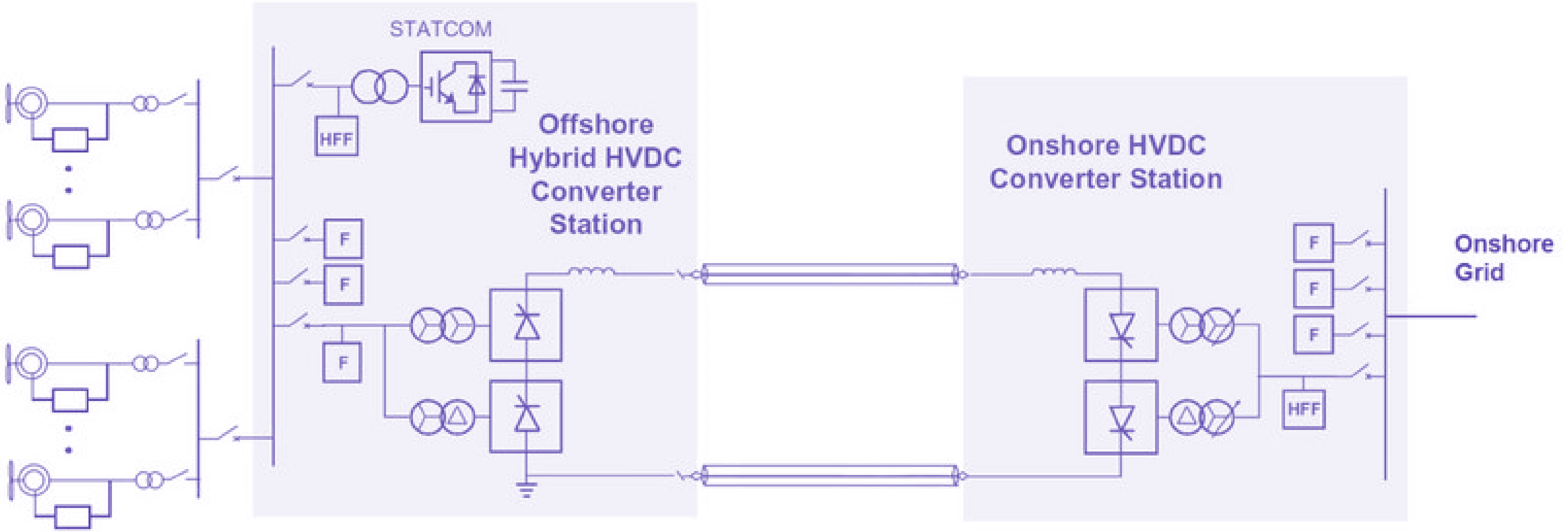
Protection Type	Formulation	Operating Mechanism	Action Strategy
Commutation Failure Protection	Y-bridge: max(IdH,IdN)−IacY>Iset D-bridge: max(IdH,IdN)−IacD>Iset Iset = max(Iset,min, Kset x max(IdH, IdN))	Detects **imbalance between DC-side and AC-side** currents using **differential principles**. Uses threshold and braking current to adjust sensitivity.	- Triggers alarms or control system switches. - Initiates emergency shutdown (ESOF) if failures persist.
Short-Circuit Protection	Y-bridge: IacY−min(IdH,IdN)>Iset D-bridge: IacD−min(IdH,IdN)>Iset Iset = max(Iset,min, Kset x min(IdH, IdN))	Monitors **AC-side current rise** and **DC-side current drop** during faults. Uses braking current and minimum action current for stability.	- Fast tripping (e.g., 0.5 ms delay). - Disconnects AC/DC circuit breakers and triggers ESOF.
Bridge Differential Protection	max(IacY,IacD)−IacY>Iset max(IacY,IacD)−IacD>Iset Stage 1: T = 200ms to reduce current Stage 2: T = 120ms to switch control system	Compares currents in Y-bridge and D-bridge connections. **Detects asymmetry** caused by faults like ground/short-circuit.	- **Multi-stage:** Control system switch (Stage 2) or ESOF (Stage 1). - Inverter-specific stages adjust for AC voltage dips.
DC Differential Protection	|IdH – IdN| > max (Isetmin, Kset (IdH + IdN)/2) T = 5ms	Detects ground faults **by comparing currents at DC high-voltage and neutral buses**. Uses ratio braking to avoid false trips.	- Triggers ESOF and isolates converters.
Valve Group Differential	max(IdH,IdN)−max(IacY,IacD)>Iset	Monitors DC-side current surge and low AC-side current during **valve faults** (e.g., misfires). Similar to bridge differential but focused on valve groups.	- Multi-stage like bridge differential. - ESOF for severe faults.
50Hz/100Hz Protection	IdL(50Hz)>Isetmin + Kset IdL IdL(100Hz)>Isetmin + Kset IdL	Detects **harmonic components** (50Hz for ground faults, 100Hz for phase faults) **in DC current**. Uses floating thresholds based on DC line current.	- **Gradual response: Alarm → Control switch → Power reduction → ESOF.** - Delays coordinated with fault recovery time.
AC/DC Overcurrent Protection	max(IacY,IacD)>Iset	Monitors **AC/DC currents** for valve temperatures. Segmented thresholds (e.g., 3.7 p.u. for **instant trip**, 1.65 p.u. for **12s delayed trip**).	- Instant trip for extreme overcurrent. - Delayed trip for sustained overloads.
AC/DC Overvoltage Protection	DC: |UdL| > Uset or |UdL – UdN| > Uset AC: $U_{AC} > U_{SET}$	**DC Overvoltage**: Detects line/neutral bus overvoltage. **AC Overvoltage**: Monitors converter voltage stress from harmonic filters and capacitors and filters 50Hz components to determine max phase voltage.	- **DC: Fast trip** (10 ms) for high overvoltage; slower for moderate. - **AC: Direct ESOF**. - **Valve Stress**: Blocks tap changes → ESOF if unresolved.
Undervoltage Protection	UdL < Uset	**DC Undervoltage**: Detects line faults or neutral bus issues. **AC Undervoltage**: Prevents prolonged low-voltage operation.	- DC: Control switch → ESOF. - AC: ESOF after delay (allows grid recovery).
Large Firing Angle Monitor	α<αset1 or α>αset2	Prevents **excessive valve stress** by limiting firing angles (e.g., >60° warning, >120° trip).	- Adjusts tap changers first. - ESOF, trip AC CB & pole isolation if angle remains critical.
Thyristor Monitoring	Count of failed thyristors > threshold	Tracks **failed thyristors via pulse signals**. Alerts at minor failures; trips at critical thresholds.	- Warning for minor damage. - ESOF for severe damage.
AC Valve Side Ground Fault	Ua+Ub+Uc>Uset	Detects ground fault with unbalance voltage during **valve block**	- Blocks converter unlock. - Trips transformer switch immediately.

Weak AC Infeed (2 < SCR < 3) with low inertia to LCC-HVDC

Weak AC Infeed (2 < SCR < 3) with **low system inertia** has an impact on operation of LCC-HVDC links.

It includes -

- Voltage Instability
- Small Signal Instability
- Commutation Failure
- Harmonics Resonance
- Limitation on Power Transfer



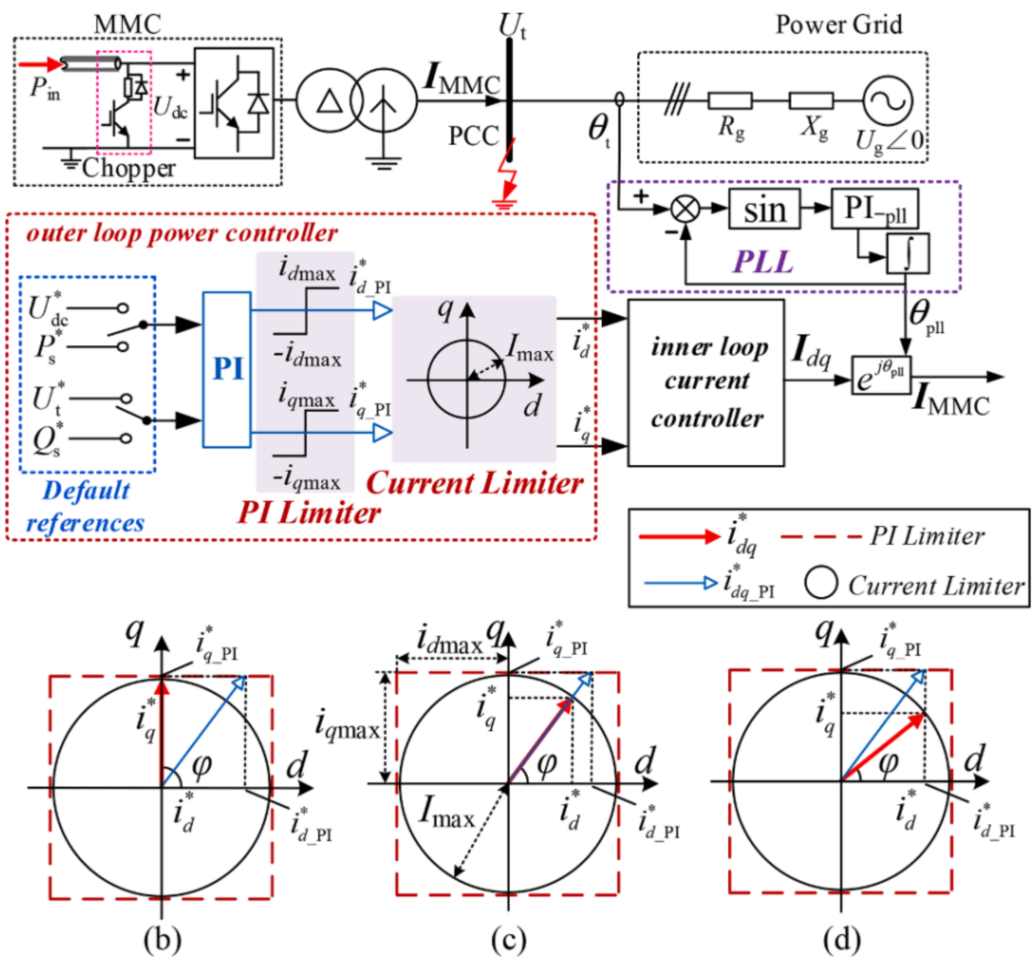
To mitigate the problems,

- Application of Capacitor Commutated Converter (CCC) with series connected capacitor to provide stable commutation voltage with less coupling to AC grid.

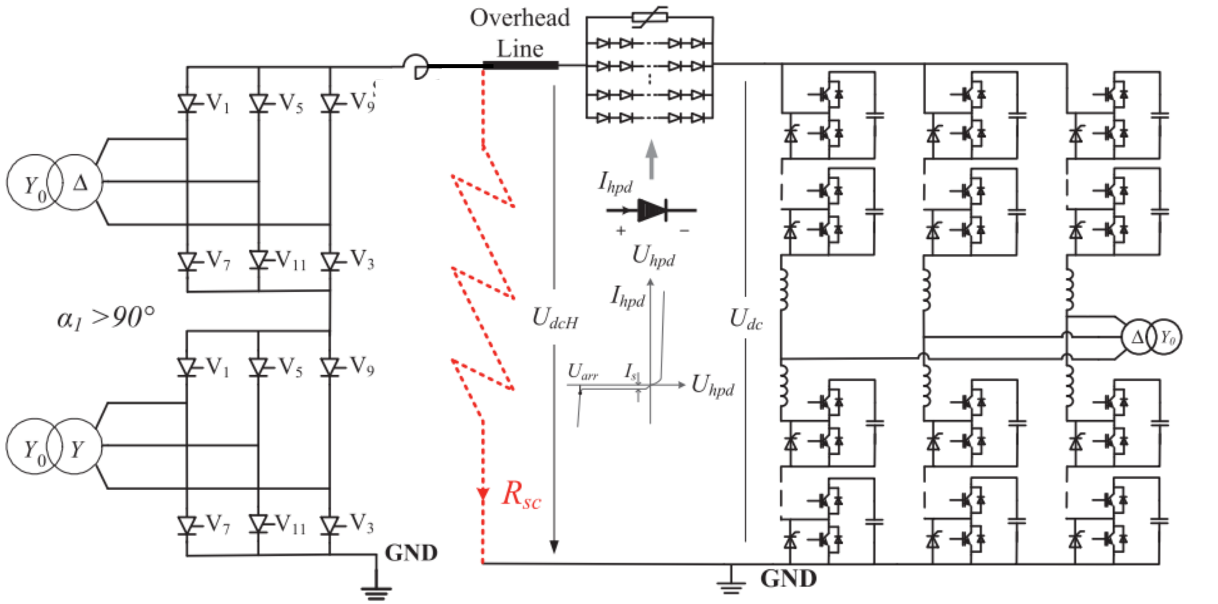
  [Note: CCC provides reactive power supply proportional to line current, hence required less reactive power compensation.]
- Application of Synchronous Condensers and VAR compensator such as STATCOM to provide voltage support

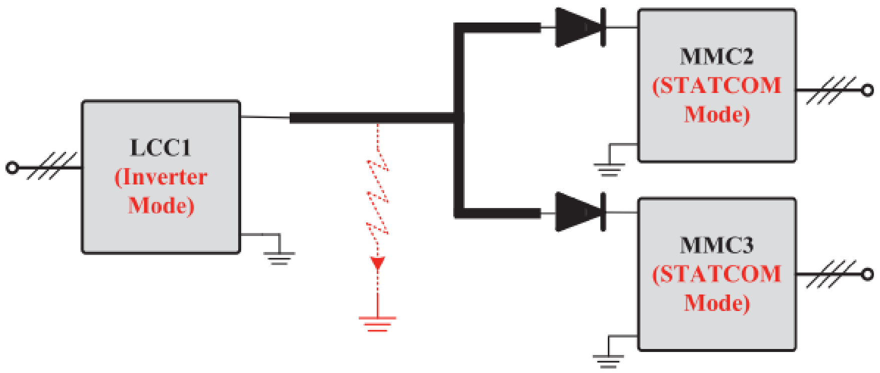
  [Note: Failure of DC Link can lead to overvoltage if NOT instantaneously tripped]

Control Structure of MMC HVDC



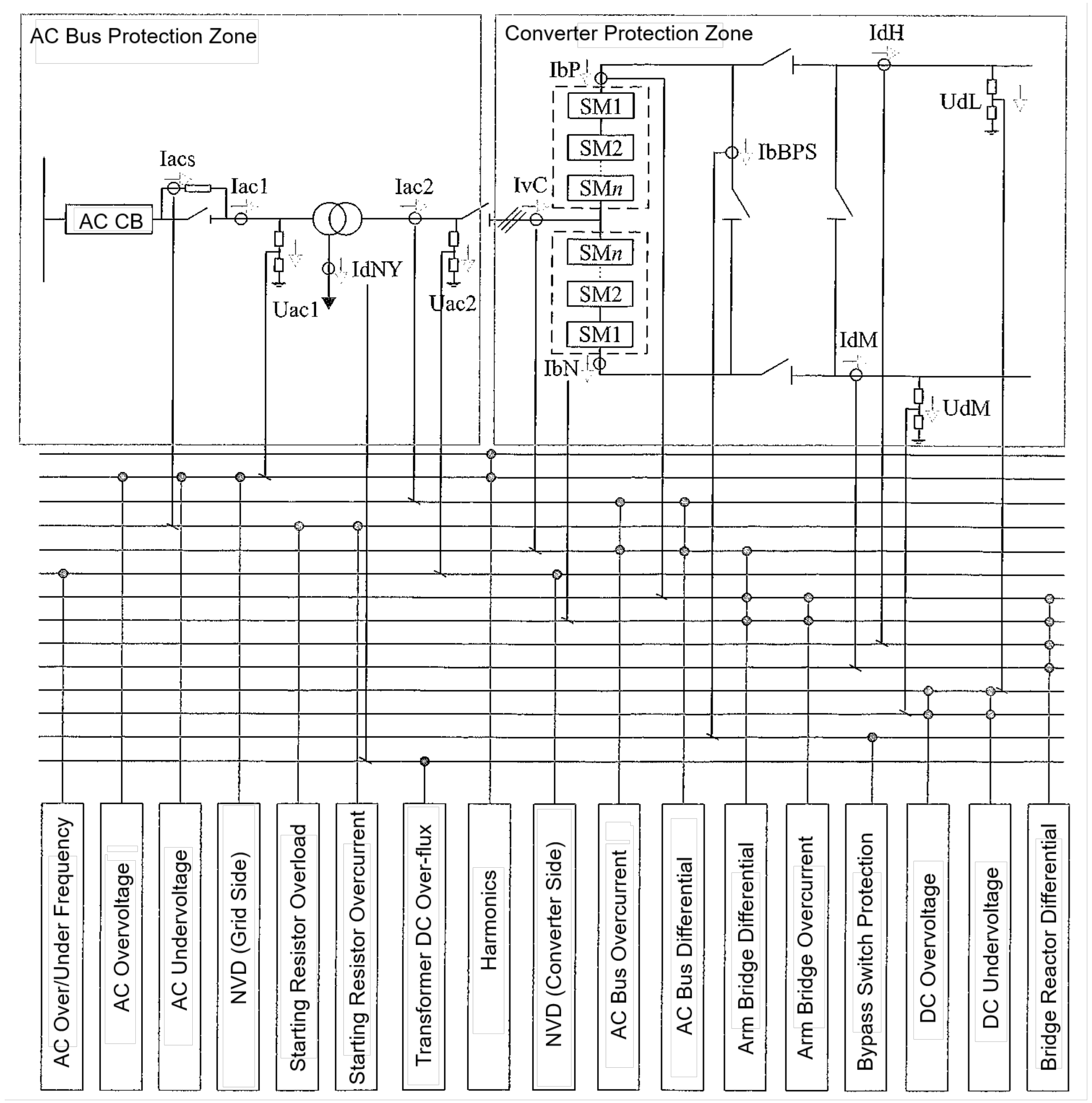
(a) The control structure of MMC,
(b) Priority limitations on active current,
(c) Limiting in equal proportion,
(d) Priority limitations on reactive current





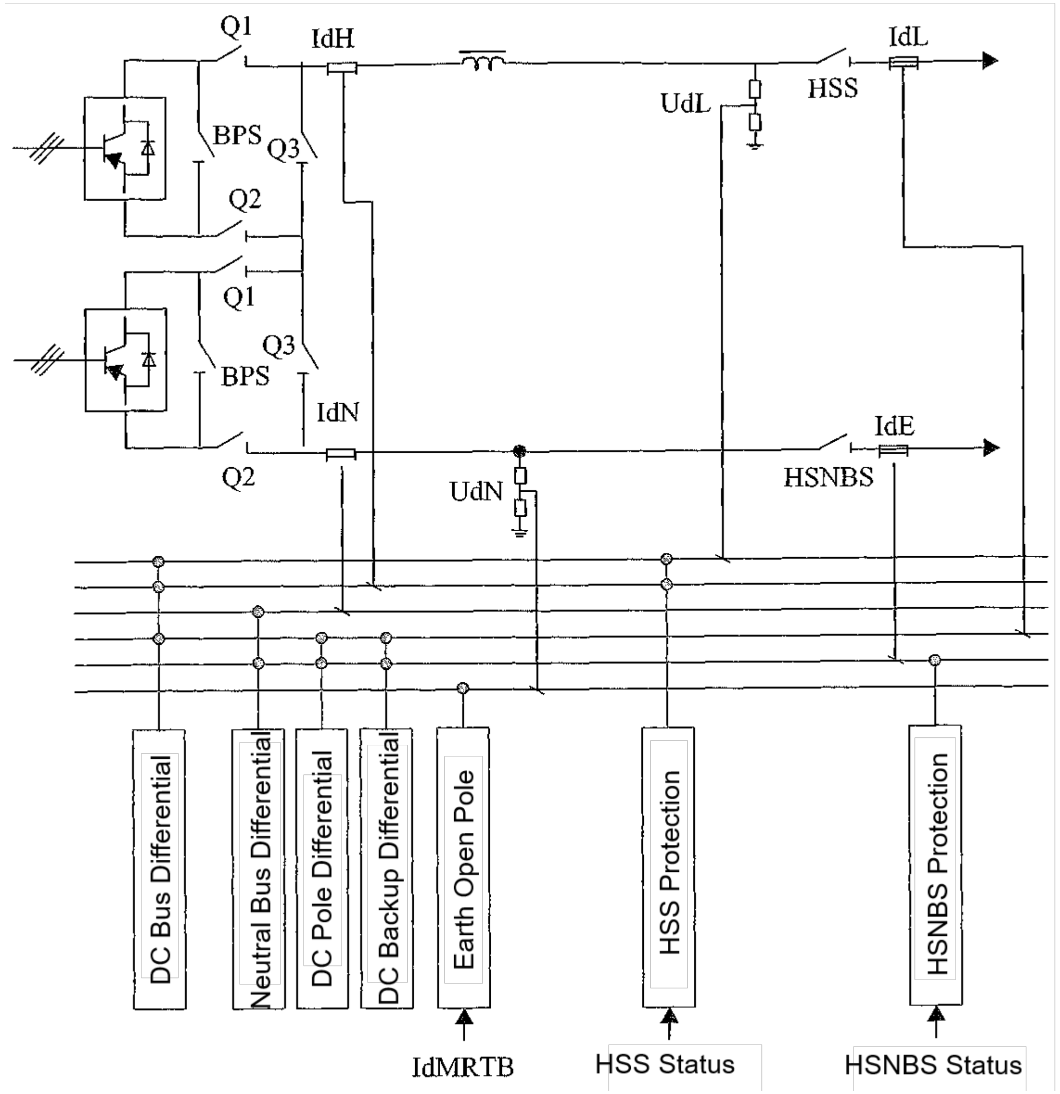
(a) HVDC Line Fault in LCC – MMC Network
(b) Equivalent Circuit and Control Mode in LCC – MMC

MMC HVDC Protection for Different Zones



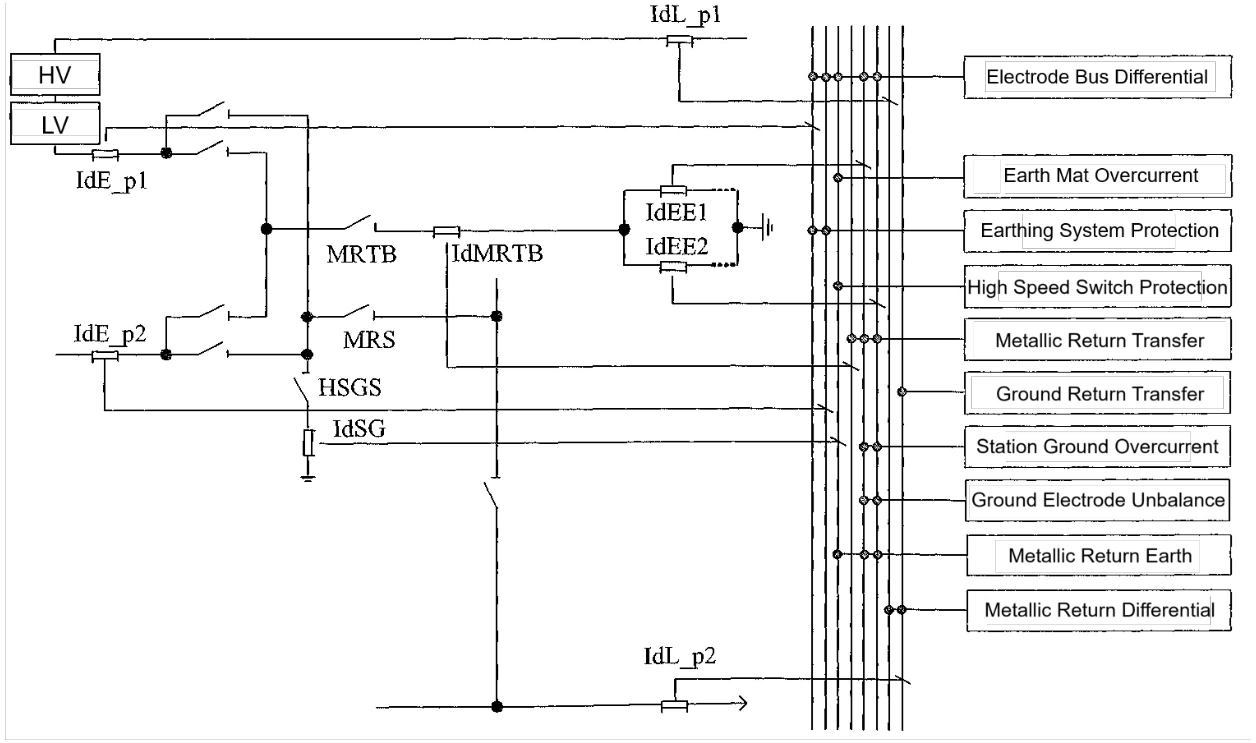
Name	Protection Task	Protection Criterion	Action Output
AC Bus Differential Protection (87CH)	Detects LL or LG faults between the converter and the flexible transformer	[Iac2 + IvC] > Iset (phase differential)	ESOF, trip AC CB
AC Bus Overcurrent Protection (50/51T)	Detects overcurrent in the AC connection bus due to faults or other causes to prevent damage to current-carrying equipment	Any of the three phases: Max (Iac2, IvC) > Iset	ESOF, trip AC CB
AC Undervoltage Protection (27AC)	Prevents abnormalities in the DC transmission system caused by excessively low AC voltage	Uac1 < Uset (using the effective value of the three-phase line voltage on the grid side)	ESOF, trip AC CB

MMC HVDC Protection for Different Zones



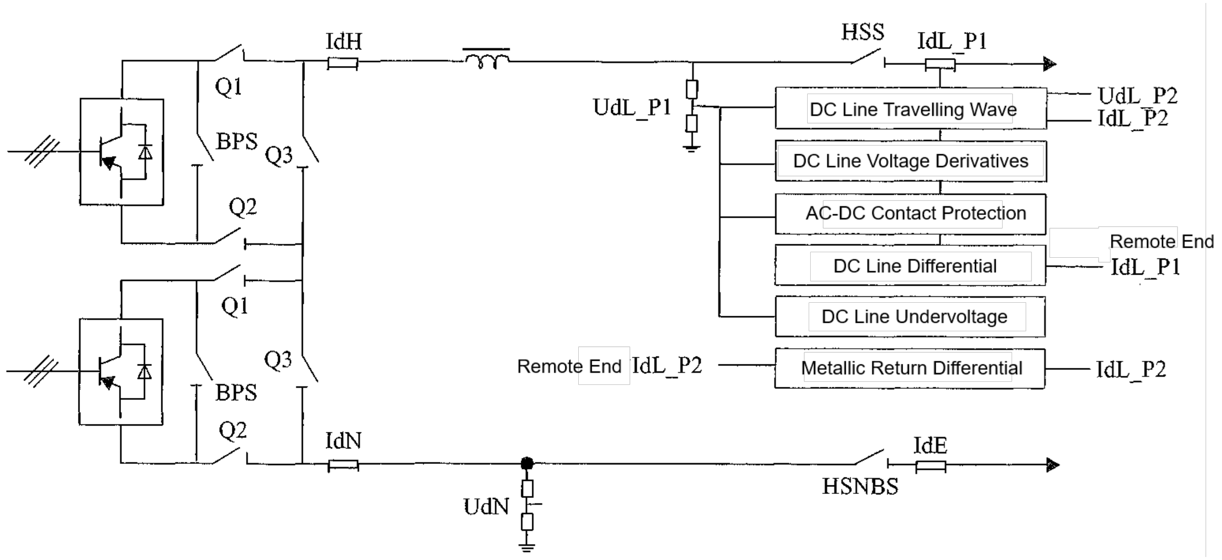
DC Bus Differential Protection (87HV)	**Stage I:** |IdH−IdL| > max (I_set, k_set × max (IdH, IdL)) & UdL < U_set. **Stage II:** |IdH−IdL|> max (I_set, k_set × max (IdH, IdL)).	ESOF, trip pole AC CB, pole isolation.
Neutral Bus Differential Protection (87LV)	**Alarm Stage: |**IdN−IdE|> I_set. **Stage I/II:** |IdN−IdE|> I_set + k_set × max (IdN, IdE).	**Stage I/II:** ESOF, trip pole AC CB, pole isolation.
DC Pole Differential Protection (87DCM)	**Alarm Stage:** |IdH−IdN|> I_set. **Stage I/II:** |IdH−IdN|> max (I_set, k_set × (IdH + IdN)/2).	**Stage I/II:** ESOF, trip pole AC CB, pole isolation.
DC Backup Differential Protection (87DCB)	**Alarm Stage:** |IdL−IdE|> I_set. **Stage I/II:** |IdL−IdE|> I_set + k_set × IdE.	**Stage I/II:** ESOF, trip pole AC CB, pole isolation.
Electrode Open-Pole Protection (59EL)	**Stage I (Bipolar Only):** UdN > U_set & |IdEE1 + IdEE2| < I_set. **Stage II (Bipolar & Single-Pole Ground):** UdN > U_set. **Stage III (Single-Pole Metallic):** UdN > U_set & IdL_op < I_set.	**Stage I:** Close HSGS, pole balancing, pole ESOF, trip pole AC circuit breaker, pole isolation. **Stage II/III:** Pole ESOF, trip pole AC circuit breaker, pole isolation.
Neutral Bus Switch Protection (82-HSNBS)	After HSNBS indicates open position, IdE > I_set.	Reclose HSNBS.
High-Speed Bypass Switch Protection (82-HSS)	**HVDC Mode:** After HSS loses closed position, IdH > I_set. **STATCOM/OLT Mode:** After HSS loses closed position, UdL> U_set & IdH < I_set.	**Stage I:** Reclose HSS. **Stage II:** Pole ESOF, trip pole AC circuit breaker, pole isolation.

MMC HVDC Protection for Different Zones



Ground Electrode Bus Differential (87EB)	**Bipolar:** $\lvert IdE - IdE\_op - IdEE1 - IdEE2 - IdSG \rvert > \max(I\_set, k\_set \times \lvert IdE - IdE\_op \rvert)$. **Single-pole Ground:** $\lvert IdE - IdEE1 - IdEE2 - IdSG \rvert > \max(I\_set, k\_set \times IdE)$. **Single-pole Metallic:** $\lvert IdE - IdL\_OP - IdEE1 - IdEE2 - IdSG \rvert > \max(I\_set, k\_set \times IdE)$.	**Bipolar:** ① Pole balance; ② ESOF, trip pole AC CB, Pole isolation. **Single-pole Ground/Metallic:** ① Reduce power; ② ESOF, trip pole AC CB, pole isolation. **No Ground:** ESOF, trip pole AC CB, pole isolation.
Ground Electrode Overcurrent (76EL)	$\lvert IdEE1 \rvert > I\_set$ or $\lvert IdEE2 \rvert > I\_set$.	**Bipolar:** ① Pole balance; ② ESOF, trip pole AC CB, pole isolation. **Single-pole Ground/Metallic:** ① Low-voltage line restart; ② ESOF, trip pole AC CB, pole isolation.
Ground Electrode Current Imbalance (60EL)	$\lvert IdEE1 - IdEE2 \rvert > I\_set$.	Same as 76EL.
Station Ground Overcurrent (76SG)	$\lvert IdSG \rvert > I\_set$.	**Bipolar & No Ground:** ① Pole balance; ② Pole ESOF, trip pole AC CB, pole isolation. **Single-pole Ground/Metallic:** ESOF, trip AC CB, pole isolation.
Metallic Return Differential (87DCLT)	$\lvert IdL - IdL1\_OP - IdL2\_OP \rvert > I\_set + k\_set \times IdL$.	Pole ESOF, trip pole AC breaker, pole isolation.
High-Speed Ground Switch (82-HSGS)	HSGS indicates open position, and $\lvert IdSG \rvert > I\_set$.	Reclose HSGS.
Metallic Return Transfer Switch (82-MRTB)	MRTB indicates open position, and $\lvert IdMRTB \rvert > I\_set$.	Reclose MRTB.
Ground Return Transfer Switch (82-MRS)	MRS indicates open position, and $\lvert IdL\_OP \rvert > I\_set$.	Reclose MRS.

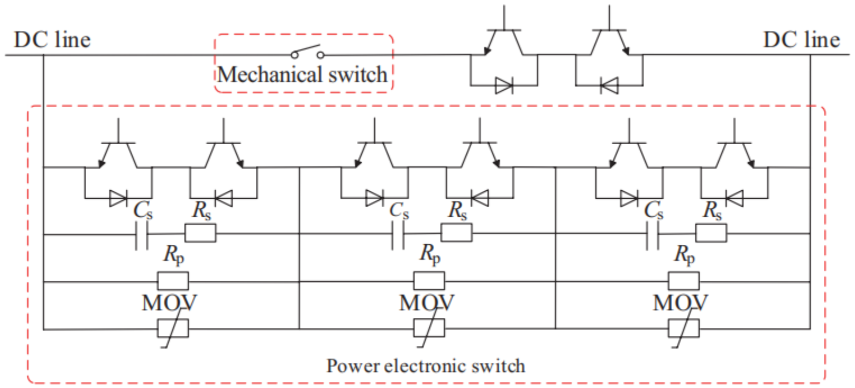
MMC HVDC Protection for Different Zones



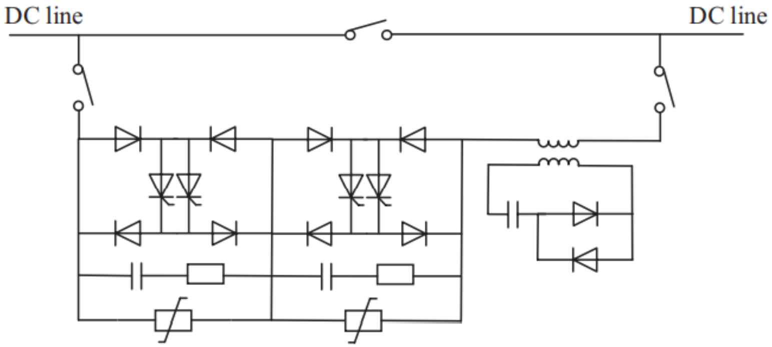
Protection Name	Protection Criteria	Action & Output
DC Line Traveling Wave (WFPDL)	Δ(Com_b(t)) > Com_dt_set & integ(Diff_b(t)) > Dif_int_set & integ(Com_b(t)) > Com_int_set	- **Station communication normal**: Line fault restart. - **Communication failure**: Pole ESOF & isolation.
DC Line Voltage Derivative (27du/dt)	Δ(UdBUS(t)) < dU_set & [UdL] < U_set	Same as WFPDL.
DC Line Undervoltage (27DCL)	[UdL] < U_set	- Station communication normal: Line fault restart. - Communication failure: Pole ESOF & isolation.
DC Line Differential (87DCLL)	IdL - IdL_os > max(I_set, k_set × IdL)	Line fault restart (disabled if station communication fails).
Metallic Return Differential (87MRL)	IdL - IdL_os > max(I_set, k_set × IdL)	Line fault restart (disabled if station communication fails).
AC-DC Contact Protection (81-I/U)	**Fast stage:** IdL > IdL_set & IdL_50Hz > IdL_50Hzset **Slow stage:** UdL_50Hz > UdL_50Hzset & IdL_50Hz > IdL_50Hzset	Pole ESOF, trip AC breaker, pole isolation.

Fault Capability for VSC – HVDC in FBSM + HBSM

Hybrid HVDC Breaker



Breaker with Magnetic Coupling



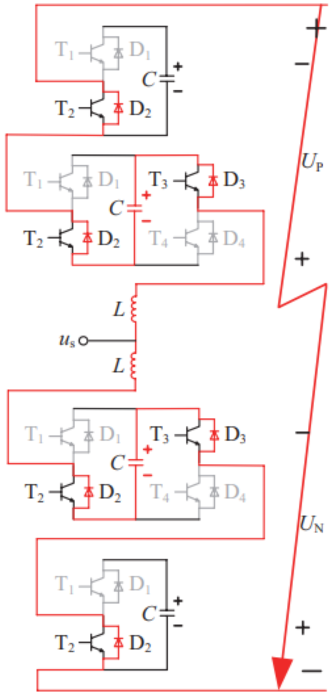


Fig. 11 Blocking current limiting operation

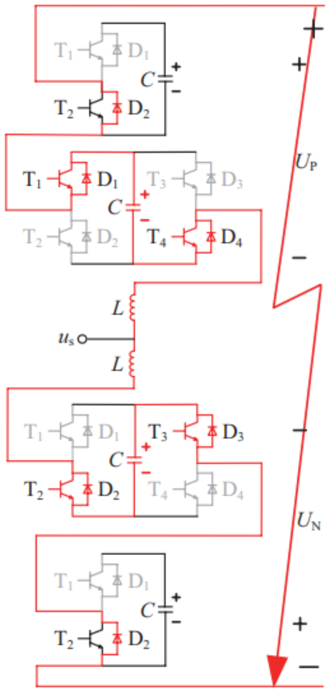


Fig. 12 Fault ride-through limiting operation

Blocking: Detecting the fault, all switching devices of SMs of the converter are blocked, so that SM capacitors are reversely connected into the fault circuit. Fault current charges the sub-module capacitors. The fault point is subjected to reverse voltage, hence blocking the DC fault current.

Fault Ride-through Mode: It controls and limits DC side current actively by reducing the DC side voltage of the converter station. It operates in STATCOM mode during fault period, which can provide reactive power support to the AC system. After the fault is isolated, the power supply can be quickly restored through this operation mode.

Fault Ride-Through Capability

Definition & Importance

- **Objective**: Maintain grid stability during DC faults by limiting fault currents, isolating faulty sections, and ensuring continuous power delivery to healthy parts of the grid.
- **Challenges**:
    - DC faults escalate rapidly due to low line impedance.
    - No natural current zero-crossing (unlike AC systems), complicating fault interruption.
    - Requires coordination with protection schemes (e.g., DC circuit breakers).

(a) Passive FRT (e.g., HB-MMC)

Mechanism:
- Blocks IGBTs during faults, diverting current to freewheeling diodes.
- Limits capacitor discharge but cannot fully interrupt fault current.

Pros: Low cost (uses standard Half-Bridge Submodules).

Cons:
- Relies on external DC circuit breakers (DCCBs) for fault clearance.
- No AC grid support during faults.

(b) Semi-Active FRT (Enhanced HB-MMC)

Mechanism:
- Adds **protective thyristors** to bypass submodules or short-circuit AC side
- Reduces fault current magnitude but lacks full control retention.

Pros:
- Lowers DCCB cost by reducing fault current.
- Faster than passive FRT.

Cons:
- Partial loss of AC grid control.
- Higher complexity than passive FRT.

(c) Active FRT (e.g., FB-MMC, Hybrid-MMC)

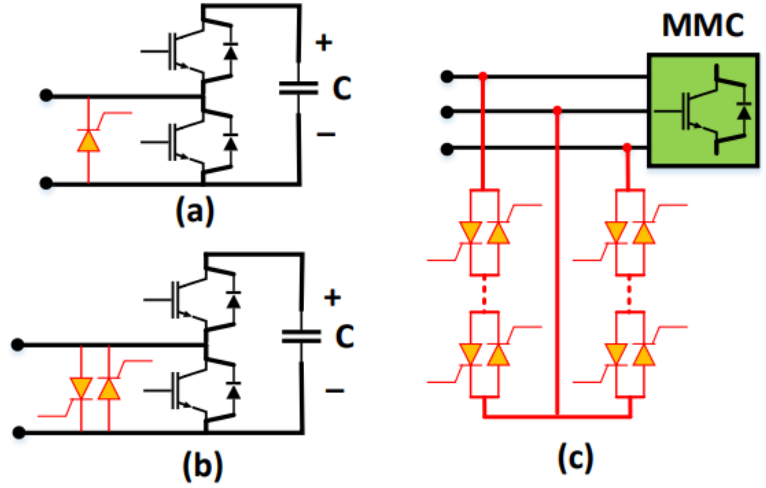
Mechanism:
- **Full-Bridge (FB) or Hybrid Submodules** generate reverse voltage to suppress fault current.
- Fully blocks fault current without external breakers.

Pros:
- Retains AC grid control during faults.
- Enables selective fault isolation.

Cons:
- Higher cost (2× power switches vs. HB-MMC).
- Increased power losses.

Fault Ride-Through Capability

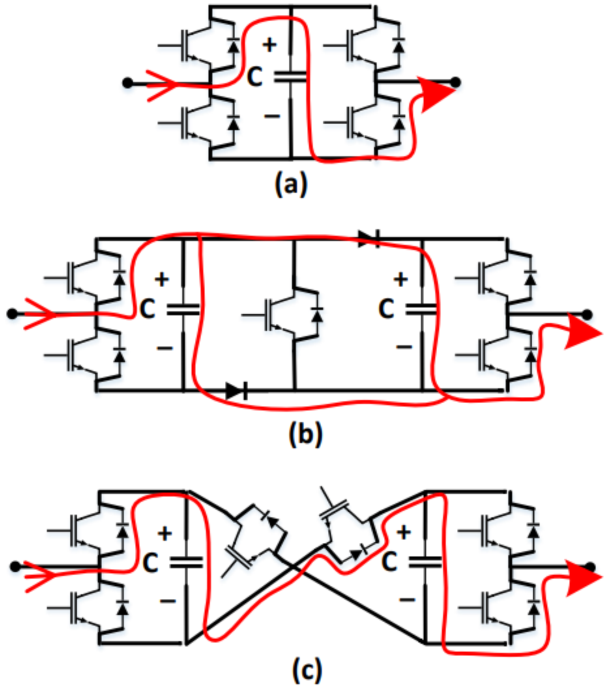
Semi-Active DC Fault Ride-Through:

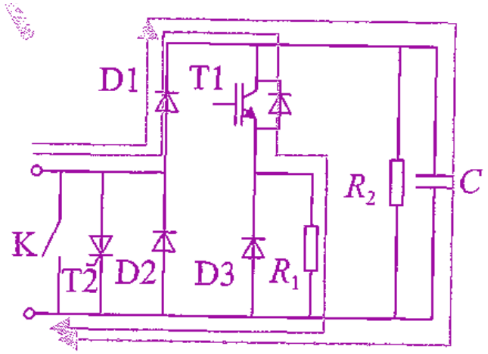


Active DC Fault Ride-Through:

Fault interruption procedure is usually performed by generation of inverse voltage to suppress fault current

(a) FB – MMC
(b) Clamped Diode Submodule (CDSM)
(c) Cross-Connect Submodule (CCSM)





Current HB Configuration:
Bypass Fault Current with Thyristor or **Short Circuit when Determined as Internal Fault**
(e.g. Overcurrent or Overvoltage)

DC FRT Class	Fault-Blocking Capability	AC Grid Support	Cost	Control Complexity	MMC Topology
Passive	No	No	Low	Low	HB
Semi-active	Yes	No	Medium	Medium	HB+Thyristor
Active	Yes	Yes	High	High	FB/Hybrid/ CDSM/CCSM

Conclusion

- Protection Philosophy in HVDC requires protection to maintain stability under different operating conditions including grid conditions. Depending on the availability of DC CB and FCL, the protection can be designed and coordinated as non-selective, partially-selective and fully-selective fault clearance.
- Compared to traditional AC systems, faults in HVDC systems are often switched fault in different modes with high rate-of-rise due to capacitor discharge. Converters should perform its additional duties (e.g. block with current reversal, bypass in submodule or arm bridge bypass instead of just detect-and-trip in AC) under controller or protection actions.
- Instead of direct fault interruption in AC CB, DC CB (divided into Passive DC CB, Active DC CB and Thyristor-based DC CB) are designed to have fault commutation to LC branch or thyristor branch to create zero-crossing or perform current chopping with aids of MOV.
- To avoid large rate-of-rise in HVDC fault, FCL is often installed. FCL should be coordinated with DCCB by bypassing the reactor after DCCB is tripped.
- For LCC-HVDC, the fault inception is limited by surge impedance and hence to be within 2.0pu and to be damped down by inverter mode reversal to possibly 0.1pu. Fault current is mainly fed by AC grid through thyristor. In AC side, it occurs as a switched phase fault. For HB MMC-HVDC, the fault develops from capacitor discharge with large rate-of-rise and circulate through LC circuit. After the submodule fault detection, the HB bypass itself with the shunt thyristor and the fault is fed from AC side through the rectifier. For FB MMC-HVDC, the fault can be ride-through with submodule capacitor reversal to damp down the fault current.
- For DC line fault, in LCC or MMC, the fault is detected with travelling wave and voltage derivative protection, to avoid the use of communication in current differential as main protection. Travelling wave approach has a fast detection (< 1ms), but the setting sensitivity is dependent on fault distance, fault resistance, fault pole, operating voltage and converter outage; voltage derivative approach has a slower response (10ms to 50ms) to avoid transient, but the setting is independent on the parameter.
- Specific protection should be provided according to the converter topology for the HVDC systems. For LCC-HVDC, the possible failure is commutation failure, arc through, misfire and current extinction. For MMC – HVDC, the detection on failure in submodule relies on its individual OC/OV detection inside the submodules, and the count on failed submodules leading to reduced voltage operation or even pole isolation.

Reference

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[2] K. Sharifabadi, L. Harnefors, H. P. Nee, S. Norrga, and R. Teodorescu, *Design, Control and Application of Modular Multilevel Converters for HVDC Transmission Systems*. Chichester, West Sussex, United Kingdom: John Wiley & Sons, Inc, 2016.

THE END

ANY QUESTIONS?

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